Coastal conservation and blue carbon: Willingness to pay for changes to nearshore management in Oregon

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Abstract

This paper reports results from a discrete choice experiment conducted with Oregon residents regarding possible policy changes in spatial management of nearshore habitat. We evaluate public preferences across several policy scenarios, each characterized by varying levels of marine reserve system size, coastal jobs generated or lost, and carbon sequestration by seagrass beds, tidal marshes and kelp forests (blue carbon habitat expansion), with models estimated in both utility and willingness to pay (WTP) space. Each of these attributes across all models displays positive, monotonic marginal WTP. Scenario analysis reveals that an "optimistic" policy package (+50 % reserve size, +200 jobs, +100 % blue carbon) commands a mean WTP of $465–704 per year, whereas equivalent contractions for reserve system size and jobs generate larger welfare losses, indicating "loss aversion" among citizens. Preference varies by recreational coastal use: non-fishing recreationists exhibit the highest WTP for attribute improvements, while recreational fishers exhibit lower WTP and would otherwise require greater compensation to accept the status quo. These findings indicate broad public support for policies that support and potentially expand blue carbon habitat, and suggest that tiered financing or carbon-credit mechanisms may be viable strategies to secure stable funding while maintaining stakeholder acceptance.
Keywords: willingness to pay; environmental policy; generalized mixed logit model; choice experiment; non-market valuation

1 Introduction

At the household level people have indicated positive preferences for improvements in coastal habitats worldwide (c.f., Doolan and Hynes, 2023; Riegel et al., 2023; Nguyen et al., 2023; and Hynes et al., 2021 for recent examples). Beyond fisheries, ecosystem services, such as carbon sequestration, have been an important element considered in marine conservation policies, yet the values associated with several of these benefits from nature remain unexplored. Marine management tends to be fishery centric, with limited attention given to non-market values associated with the ocean (Smith and Willen, 2003; Albers et al. 2020; Albers, 2022). Valuation of potential environmental outcomes, particularly carbon capture and storage (blue carbon) arising from management of kelp forest, seagrass forests, and tidal marshes are crucial for understanding the full benefits of managing marine environments.

Policy commitments and legal frameworks for area-based marine conservation have intensified in ways directly relevant to Oregon's marine reserve system (MRS). In the United States, Executive Order 14008 launched the America the Beautiful initiative and set a national goal to conserve at least 30% of U.S. lands and waters by 2030 (Exec.

Order No. 14,008, 2021; U.S. Department of the Interior et al., 2021). The U.S. Department of the Interior (DOI), outlined implementation steps in the 2021 report and subsequent annual updates. Internationally, the U.N. High Seas Treaty reached the ratification threshold on September 19, 2025 (United Nations, 2025), enabling it to enter into international law and explicitly aiming to place 30% of the high seas under some protected status by 2030 (Convention on Biological Diversity, 2022). Together, these national and global commitments increase the demand for credible evidence on how the public trades off cost and benefits from spatial management of nearshore habitats.

Coastal and marine habitats, covering over 70% of the Earth's surface, are increasingly recognized for their role in biodiversity preservation and carbon storage (Siegal, D. et al, 2021). Public-perception research finds that "nature-based" carbon dioxide removal options, e.g., maintaining and/or expanding tidal estuaries and sea grass habitats, are viewed more favorably than "engineering-based" interventions, e.g., direct injection of liquid CO2 into the seabed, iron fertilization, and alkalinity enhancement (Nawaz et al., 2023; Betram and Mark, 2020). A partial explanation for this increased acceptability is that, compared to engineered options, nature-based interventions not only provide carbon mitigation, but also broader co-benefits such as biodiversity conservation, expansion of fish nursery habitat, and job creation opportunities.

This paper evaluates preferences related to spatial management of nearshore habitat within Oregon's territorial sea (high tide line to 3 nautical miles (5.6 km) offshore) and associated job creation opportunities. We incorporate several innovations in a discrete choice experiment (DCE) designed to estimate heterogeneous preferences among survey participants. Policymakers are often unaware of the extent to which people value changes in marine conservation strategies, and are even less informed of non-market, or passive-use values, as opposed to market values (such as the value of fish harvesting). Providing robust non-market valuation is crucial to assessing household's policy acceptance and to properly design financing mechanisms that could support, for example, expanding nature-based carbon sequestration.

Evaluating climate change mitigation benefits along with changes in marine reserve size and job creation yields a broader understanding of relevant tradeoffs involved in nearshore spatial management policies, and supplies additional non-market value estimates needed for credible net social benefit or cost calculations. Furthermore, because different discrete choice models capture behavioral heterogeneity in distinct ways, they can produce inaccurate and unreliable welfare estimates for nearshore spatial management scenarios (Train and Weeks, 2005; Daly et al., 2012; Thiene and Scarpa, 2009). Accordingly, this research assesses alternative model specifications, selecting those that most accurately reflect choice behavior and public acceptance of possible changes to Oregon's nearshore spatial management policy.

Relatively few studies have examined public valuations of coastal habitat management

(Doolan and Hynes, 2023). Most prior research has focused on assessing the public's willingness to pay (WTP) for habitat expansion and management (i.e., seagrass or kelp restoration) rather than valuing the specific ecosystem services generated (i.e., carbon sequestered)(Hynes et al., 2021; Aanesen et al. , 2015; Grilli et al. 2022; Kim et al. 2022). Neither have they tested a broad array of models/specifications to rigorously evaluate the effects of distinct types of heterogeneity on choice behavior (Riegel et al., 2023; Tyllianakis, 2022). Existing work on blue carbon tends to employ three choice modeling approaches - Multinomial Logit (MNL), Random Parameters Logit (RPL) and Latent Class Models (LCM) - controlling solely for taste heterogeneity at the individual level. These studies confirm positive and significant WTP estimates for blue carbon, and reveal substantial taste heterogeneity where LCM models isolate a relatively small portion of the sample that evaluates this attribute negatively (Riegel et al., 2023; Tyllianakis, 2022).

Early research on protecting coastal forest ranges in Oregon explored public preferences during the initial implementation phase of protection (Garber Younts et al. 2004). Later work assessed support for changes in reserve size, but did not account for blue carbon or possible employment effects linked to its management (Lindberg et al. 2019). Oregonians have shown increasing support for the state's marine reserves over time, with results indicating that favorability is strongest among individuals living near the reserves and in the state's more densely populated regions (McAlpine et al., 2025).

Understanding heterogeneous preferences among recreational ocean users can inform marine policy processes and outcomes (Johnston et al. 2010; Arlinghaus et al. 2019). For example, because recreationists (individuals who recreate in a variety of different ways, e.g., sightseeing, beachcombing, etc.) often value reserves beyond biomass protection, their incentives may diverge sharply from those of commercial fishers, implying different optimal management strategies from a theoretical perspective. Sustainable management actions can do more than avert ecological damage because they can also create benefits associated with the development of tourism infrastructure. Choice experiment evidence shows substantial WTP values for recreation and habitat protection (Davis et al.,2019), for tourism enhancements such as more boat docks, hiking trails, etc. (Paltriguera et al., 2018), and avoidance of illegal fishing (Hall et al., 2002). In our study, we stratify respondents into two subsamples according to the frequency of their ocean recreation activity in order to examine how WTP for reserve attributes varies across these two groups. This segmentation allows us to estimate policy implications corresponding to different user profiles, ensuring that management decisions have the potential to consider the heterogeneity of public preferences.

Past research in various contexts have estimated welfare gains associated with protecting and/or expanding different environmental amenities associated with marine ecosystems. However, valuation of blue carbon is a relatively novel research topic. We ask: How

much are Oregon residents willing to pay for coastal management policies that expand or contract reserve size, impact coastal employment, and affect blue carbon levels? How do these valuations vary across individuals who actively use marine areas for recreation and those who do not? What models best capture Oregonians' choice behavior and provide more accurate estimates than those reported in previous studies?

To explore these questions and quantify how people value different marine policy outcomes, we conducted a DCE with 371 Oregon residents located across the state. The experiment was designed to estimate public preferences, in particular how individuals evaluate trade-offs among three key attributes: (1) changes in marine reserve size; (2) changes in coastal employment – across fisheries, tourism and blue carbon management; and (3) changes in blue carbon (carbon sequestered by seagrass, estuary and kelp habitat). We ran several types of choice models to assess household valuation of these attributes across distinct scenarios, and what types of heterogeneity affect households' choices regarding changes in marine policy parameters. We evaluate the effects of taste heterogeneity, scale heterogeneity, shared unobserved similarities among policy alternatives (non-status quo), and choice uncertainty on household preferences.

Toward these ends, we ran Random Parameter Logit (RPL), Random Parameter Logit with Error Components (RPL-EC), and Generalized Mixed Logit models in utility and WTP space. Estimating our model in WTP space provides an essential comparison with the models estimated in utility space in terms of the distributions of individual WTP estimates (Cerroni et al., 2023; Chen et al., 2023; Hess and Train, 2017). Also, we specify the models in linear and non-linear forms to capture both unit and aggregate utility coefficients. Models with interactions between attributes are also estimated to capture trade-offs among them. We select the best linear model by comparing goodness-of-fit measures, and perform scenario analyses taking the posterior individual WTPs from the models estimated in WTP space, and interpreting the associated WTP distributions as inverse demand functions for the distinct scenarios.

To determine whether Oregonians value marine reserve attributes differently based upon their recreational habits, we divided the sample into two subgroups – one comprised of recreational fishers, the other comprised of non-fishers. To test whether the WTP distributions are statistically different across these two subgroups, we performed comparisons using Poe's convolutions and Wilcoxon median tests. We also performed statistical difference tests between marginal WTPs derived from models that account for the effects of choice uncertainty on respondents' preferences. Following Börger et al. (2021) we tested for attribute non-attendance (ANA) of the status quo itself, rather than of a given attribute, such as an alternative's overall cost à la Lew (2019) [1].

[1]We estimated the existence of potential latent ANA among our respondents, unlike Breeze et al. (2015), who rely on respondents self-identifying their non-attendance. An interesting line of research, somewhat related to ANA, is what Howard et al. (2020) label "attribute value substitution", whereby respondents substitute what they believe are more reasonable attribute levels for those not presented in

We find that increases in marine reserve size, coastal employment, and blue carbon are each associated with positive marginal WTP estimates. Annual WTP for (1) expanding the size of the MRS ranges from \$2.05 to \$3.57 per percent increase, (2) increasing blue carbon ranges from \$1.72 to \$3.30 per percent increase, and (3) increasing the number of coastal jobs ranges from \$0.67 to \$0.98 per job. All else equal, WTP to avoid the status quo ranges from \$110.56 to \$238.46 per year.[2]

Through estimation of a non-linear form of our model, we find strong evidence of loss aversion among survey participants. For instance, the effect on utility of a 50% decrease in the size of marine reserves is noticeably larger in magnitude than the effect of a 50% increase in size, translating to respective annual WTP values of -\$224 and slightly less than \$124. Even more pronounced is the relationship between decreases and increases in the number of jobs, where the magnitude of loss aversion is almost \$150, going from -\$149 for the decrease in jobs to a statistically nonsignificant \$46. A host of additional empirical results addressing the issues of scenario analysis, tradeoffs among the different attributes, preference heterogeneity, and the existence of ANA are also presented.

Respondents tend to tradeoff jobs and reserve area for more abundant blue carbon, and become less price sensitive at high blue-carbon levels, signaling a premium for carbon-efficient coastal management scenarios. Heterogeneity matters, and capturing correlation in unobserved utility among "no-status quo" alternatives along with distinct individual tastes reveal more nuanced choice patterns among respondents. Recreational fishers show lower marginal WTP for all attributes, but demand more compensation for retaining the "status quo" than non-fishers. In general, those individuals who have recreated along the coast in the past (recreationists) exhibit consistently higher WTP for blue carbon than those who have not (non-recreationists).

The remainder of this paper is organized as follows. Section 2 provides a brief review of the literature. Section 3 describes the Oregon nearshore spatial management. Section 4 presents the data along with an array of summary statistics. Section 5 outlines the methods and estimation strategies employed within the random utility framework. Section 6 presents our empirical results. Section 7 places the empirical results in a broader perspective, and points out their limitations. Section 8 offers conclusions and suggestions for future research.

2 Literature Review

Several recent studies establish a baseline for our empirical results concerning household preferences for key attributes of nearshore management. A handful of additional

the scenarios, and then answer accordingly.

[2]Breeze et al. (2015) interpret this latter estimate as an existence value (the innate value respondents attach to knowing that a policy is in place).

studies serve as methodological guideposts for this analysis to obtain accurate estimates of these preferences. This section briefly summarizes each of these studies in order to situate our theoretical framework and attendant findings in the context of a broader literature.

2.1 Baseline for Empirical Results

Lindberg et al. (2019) assessed the preferences of Oregonians for expanding or contracting the state's marine reserves designated in 2012, and attendant changes in employment opportunities in the coastal region. Preference heterogeneity, was assessed relative to respondents' environmental worldviews, recreational usage of the reserves, and whether they work in the commercial fishing industryThe authors find that annual household WTP (in 2024 dollars) for a 1% increase in the state's marine reserve area is roughly -$4.60, when the respondent for that household identifies as highly anthropocentric, not well-connected to nature, never engages in coastal or ocean recreation but does so with recreational fishing, and works in the commercial fishing sector. In contrast, annual WTP jumps to roughly $9.35 for a respondent who identifies as low anthropocentrism, well-connected to nature, often engages in coastal or ocean recreation and not in recreational fishing, and does not work in the commercial fishing sector..

Prior to Lindberg et al. (2019), Garber-Yonts et al. (2004) assessed household WTP for inter alia three attributes related to the Oregon coastal range: (1) percent of Oregon's coastal zone designated as biodiversity reserves, (2) percent of endangered species habitat in the coastal range in which species are protected at the highest protection standard, and (3) percent of coastal salmon habitat protected at highest protection standard. They estimate state-weighted mean annual household WTP values for 10% increases in (1) biodiversity reserves of $342, (2) endangered species habitat of $223, and (3) protected salmon habitat of $102. Interestingly, the authors find that respondents are willing to pay roughly $294 to avoid any of the alternatives that increase or decrease biodiversity conservation, regardless of cost or the degree of increase or decrease in the extent of the change).

Focusing on the restoration of coastal sand dune ecosystems in the US Pacific Northwest (PNW), Nguyen et al. (2023) find that respondents prefer larger restored areas, full restoration, and no change to recreation access. While WTP increases resulting from expanding the size of restored areas are modest, substantial gains in WTP are realized from increasing the quality of restoration. While changing recreation access generally results in disutility, programs that restrict recreation may still yield positive WTP if high-quality restoration quality is achieved.

In their study of preferences for kelp forest restoration among the Norwegian public, Hynes et al. (2021) estimate both conditional and mixed logit models and obtain ranges

of annual WTP values for enhanced biodiversity levels (between roughly $23 and $38) and juvenile fish nurseries (between $17.70 and $24.50) depending upon abundance of species, as well as for increased area restored (between $8 and $36.70) for 10,000 m2 and 40,000 m2, respectively. These values are higher among respondents who are active marine environment users. The authors estimate a compensating surplus of between $84.30 and $100.60 for a scenario with high abundance for both biodiversity and juvenile fish nurseries and a 40,000 m2 area of restoration. Meyerhoff et al. (2021) likewise estimate household preferences for coastal adaptations along the German Baltic Sea, an area characterized by brackish water bodies forming lagoons. The authors estimate average annual WTPs for increased beach nourishment of over $13.60, and dyke realignment to mitigate the potential effects of climate change of between $15 and $24.50. Regarding blue carbon, studies show that tidal marshes provide habitat with the highest potential for carbon sequestration by area (Lyle et al. 2022).

Riegel et al. (2023) estimate WTP for carbon-storage services provided by Scotland's saltmarsh ecosystems. The authors find a non-monotonic annual WTP response to increasing levels of blue carbon, first increasing from roughly $92 for a 1% increase in blue carbon to $114 for a 2.5% increase, then decreasing to $67 for a 5% increase. Respondents provided with more scientific information about the saltmarshes did not necessarily make more coherent choices (in terms of variability in choice outcomes) than those who were not provided with such information.

Lastly, in their study of respondents in British Columbia, Canada, and Washington, US, Nawaz et al. (2023) find that respondents were most comfortable with enhancing blue carbon via coastal restoration and least comfortable with ocean-alkalinity enhancement. Among those respondents with greater odds of expressing climate severity and urgency, the odds of being comfortable with any type of blue carbon (ocean alkalinization, ocean fertilization, direct off-shore carbon capture, or coastal restoration) are higher, though still highest for coastal restoration. Interestingly, a respondent's proximity to the coast did not influence their preferences for blue carbon.

2.2 Baseline for Analysis

Several recent studies point the way in terms of accounting for preference heterogeneity associated with sociodemographic and intrinsic attitudinal characteristics, exemplified by (1) Abate et al. (2020), who find that respondents who are more concerned about marine plastic pollution in general are WTP more than respondents who are not, (2) Aanasen et al. (2023), who find no evidence supporting the hypothesis that respondents supplied with information about the uncertainty of the negative effects of aquaculture on Arctic Norway's marine environment are WTP less than respondents who were not supplied with this extra information, (3) Lindberg et al. (2019), who find that preferences for

expanding marine reserves in Oregon depend heavily on environmental worldview, in particular degree of anthropocentrism and connectedness to nature, as well as level of engagement in coast-orientated recreation and recreational fishing, and (4) Li and Ando (2023), who find that values for grassland restoration in the US Midwest are especially high for those who could have developed a place identity for grasslands by living within walking distance of one as a child, and for those who had immersive nature experiences like camping and environmental education. We employ both the subsampling approach of these studies, as well as the estimation of the effects of these sociodemographic and attitudinal characteristics on individual-level marginal WTPs for each attribute derived from the sequence of observed choices in the experiment (c.f., Chen et al, 2023, and Li and Ando, 2023).

Following Chen et al. (2023), we account not only for preference heterogeneity based upon sociodemographic and attitudinal characteristics, but also the underlying effects of scale heterogeneity and heterogeneity associated with the alternatives themselves. To accomplish this, we estimate both mixed and generalized mixed logit models, with and without error correction (Section 5 provides an in-depth discussion of the theoretical foundations of these models). Lastly, our analysis incorporates several innovations that have been developed in the recent literature, including tests of the face validity of WTP estimates (Glenk et al., 2024), loss aversion (Dugstad et al., 2021), and WTP to avoid reductions in attribute levels from status quo levels (as opposed to solely WTP for increases in attribute levels from status quo) (Breeze et al., 2015). These and other innovations are investigated in Section 5.

3 Study Area

Oregon's territorial sea extends three nautical miles (5.6 km) offshore. The state's coastline length is approximately 362 miles (583 km), spanning eight coastal counties. Statewide planning for marine reserves began in the late 2000s. The Ocean Policy Advisory Council (OPAC), as directed by Executive Order 08-07, stipulated the goals, implementation principles and guidelines, and objectives for the creation of Oregon's marine reserve system, initially restricting the number of sites to no more than nine (Oregon Office of the Governor, 2008). In 2009, House Bill 3013 (Oregon Legislative Assembly; 2009) created the marine reserve program and funded a pilot phase at Redfish Rocks and Otter Rock sites. Senate Bill 1510 (Oregon Legislative Assembly; 2012) defined a work plan to implement additional reserves from 2012 to 2023.

The state's marine reserve system (MRS) encompansses both protected areas, where some extractive uses are allowed, and marine reserves, where extraction and development are prohibited. In addition, no active management beyond monitoring is allowed in marine reserves.

The five no-take marine reserves account for approximately 3% of Oregon's nearshore habitat, while the marine protected areas account for an additional 6%. These two classifications are hereafter combined, for a system area of 9%, and referred to as the marine reserve system. The system, combining marine protected areas and marine reserves, is the focus of one attribute in this DCE. Figure 1 displays the locations of the specific marine reserves along the Oregon coast.

The primary goals of the marine reserve system are to enhance biodiversity and habitat conservation, serve as scientific reference sites, and avoid significant adverse impacts to ocean users and coastal communities (ODFW, 2022). ODFW is tasked with overseeing scientific monitoring, public engagement, rule enforcement (in partnership with the state police), and conducting five-year reviews. The legislation required a comprehensive evaluation after ten years, which was delivered in 2023-24, to guide potential adaptive management adjustments.

[INSERT FIGURE 1 HERE]

With respect to employment along the Oregon coast, approximately 1250 people currently work in commercial fishing, and approximately 13,500 are employed in jobs attributed to tourism (our estimate based upon SOED (2024) and Ghahramani (2020). These figures reflect the significant role that coastal tourism plays in regional employment and income, underscoring the importance of incorporating non-market values such as recreation and ecosystem services into coastal management objectives.

4 Methods and data

We begin this section with a discussion of the survey instrument and embedded choice experiment designed to elicit household preferences for different marine reserve scenarios. We then discuss the data and associated summary statistics.

4.1 Survey Instrument and Choice Experiment

The survey instrument developed for this study uses a standard approach to understand Oregonian's preferences for changes in the nearshore spatial management (c.f., Nguyen et al. (2023), Riegel et al. (2023), and Chen et al. (2023) for examples of similar survey approaches). The survey design follows guidelines developed by Johnston et al. (2017)[3]. To ensure the capture of key dimensions and the elicitation of a wide range of values associated with Oregon's nearshore spatial management, we carefully designed and operationalized the questions, and incorporated feedback from experts and stakeholders through three focus groups[4]. We then conducted a pilot evaluation of the

[3]Our survey instrument is available upon request from the corresponding author.

[4]Feedback from these groups on a draft of the survey helped guide the research team in clarifying the survey's introductory materials. For example, focus group participants were generally unfamiliar with

survey instrument, created in Qualtrics XM, in June and July 2023, first informally with graduate students and professors at Utah State University, and later with a subset of 100 survey panel participants residing in Oregon and recruited by Qualtrics. Survey content was revised based upon these pilots. The final version of the survey was administered between August and December of 2023 to another 373 survey panel participants, and an additional 68 respondents recruited via mailed invitations from a sample frame provided by Dynata, Inc., following the Tailored Design Method (Dillman et al., 2014)[5].

The survey starts with an informed consent letter followed by questions regarding the participant's use of and familiarity with Oregon's coast. Next, we provide background information on the reserves, which included the Figure 1 map and Figure 2: Panel A graphic, as well as an enumeration of the benefits and costs associated with preserving, monitoring, and administering the reserves, including the enforcement of regulations[6].

Because it addresses the issue of blue carbon within a coastal ecosystem framework – an issue that is likely novel to many of our participants – the survey provided the following information to participants,

> *"Currently, human activities release more carbon into the atmosphere than they capture and store. Reducing the risks of extreme weather, sea level rise and wildfires requires capturing and storing carbon dioxide. Marine and coastal ecosystems including kelp forests, seagrass beds, and tidal marshes can remove carbon from the atmosphere. These ecosystems can capture and store more carbon per area than most types of land-based forests. Restoring and maintaining these ecosystems could help reduce carbon dioxide in the atmosphere."*

To gauge survey participant views on climate change, we utilized a modified climate urgency scale based on Nawaz et al., 2023.

Two attention checks were utilized for quality control. Following Oh and Gramig (2023), participants who answered either of these two checks incorrectly were excluded

the terminology "carbon sequestration", which was therefore replaced with "capture and store carbon" in the survey's final version.

[5]The first round of testing incorporated cognitive interviews with potential participants, as recommended by Johnston et al. (2017). We altered the cost interval and the blue carbon levels after the 100 piloted surveys, and made no changes to the survey design thereafter. As a result, the 100 piloted surveys were discarded. Our survey pretesting procedure follows the approaches used in Wallmo and Edwards (2008), Jobstvogt et al. (2014), Chen et al. (2023), Meles et al. (2023), Aanesen et al. (2023), and O'Dell et al. (2025).

[6]Benefits are listed as: (1) protection and restoration of habitat and species, (2) scientific research, educational, and tourism opportunities, and (3) potential fisheries enhancement, such as by protecting fish nursery grounds. Costs are listed as: (1) reduced access for commercial and recreational fishing, with possible displacement to other areas, (2) financial costs of administering the nearshore spatial management, including monitoring human activities and enforcement of regulations, e.g., only specific types of fishing gear are allowed in certain MPAs, and (3) financial costs of ecological monitoring, e.g., scientists track how habitat and fish populations change over time in the nearshore spatial management.

from analysis[7].

The choice experiment was presented next, followed by a series of statements gauging the participants' (1) environmental worldviews, including statements like, "Nature is important mostly because of what it can contribute to human well-being," (choose from "strongly disagree" to "strongly agree"), (2) social values, with statements such as, "Freedom – liberty for everyone to think and act as they consider appropriate," (choose from "not important" to "extremely important"), and (3) individual values, with statements such as, "Tradition is important to [me]. [I] try to follow the customs handed down by [my] religion or [my] family," (choose from "not like me at all" to "very much like me").[8] Standard socio-demographic questions comprised the final section of the survey.

We developed an MNL d-efficient fractional factorial design for the choice experiment (which corresponds to a 3x3x4x7 full factorial design matrix, refer to Table 1 below) using the Ngene software package, with pilot design priors based upon our previous study and final design priors based upon pilot results[9]. The design includes 16 choice tasks divided into four blocks, with each respondent completing four choice tasks in randomly-assigned blocks. The design's d-efficiency score after eight iterations on the final experimental design was 1.373.

Each choice situation consists of two separate policy-change alternatives (alternatives 1 and 2) and an "opt-out" status quo option. Each of the alternatives is comprised of four separate attributes, with Table 1 providing their levels and variable names for ensuing empirical analysis.

[INSERT TABLE 1 HERE]

The attribute *MR* corresponds to percent changes in the size of the marine reserve system. *Jobs* refer to changes in the number of jobs related to fishing, blue carbon, and tourism along the Oregon coast. Local and state government subsidies for attracting and retaining businesses[10] reflect public support for job creation in their communities. The present focus is not on the social welfare effect of incentive programs, but on public values for job creation. Research indicates positive resident WTP for jobs, including in coastal and marine contexts (Xuan et al. 2021).

Carbon reflects the percentage change in carbon sequestration relative to the status

[7]Scarpa et al. (2009) solely removed respondents who chose only the status quo. Riegel et al. (2023) included three "red herring" questions to test whether respondents read the provided information text; respondents were screened out if they replied incorrectly to two out of the three questions. Respondents were also screened out due to speeding concerns, as well as via failing an attention test similar to ours. This is in keeping with Malone and Lusk's (2018) "trap questions," which led to the conclusion that "participant inattention can substantially bias parameter estimates and subsequent policy recommendations (page 219)".

[8]Each question also included "not sure" as a choice option. Individuals who answered this alternative were considered as "NAs" and excluded if the question was included as a variable of analysis.

[9]For information on Ngene see https://www.choice-metrics.com.

[10]See, for example, https://www.mckinsey.com/industries/public-sector/our-insights/how-state-and-local-governments-win-at-attracting-companies/

quo. This metric is consistent with other studies (Bocci et al. 2020; Brodersen et al. 2024) and a reflection of the complexity and uncertainty in calculating carbon sequestration (Börger et al. 2014; Luisetti et al. 2014; Macreadie et al. 2019). The percentage change metric also reflects the project goal of providing an initial assessment of public preferences, rather than an estimate of WTP per ton of carbon dioxide equivalent for use in benefit-cost analysis.

Lastly, Cost denotes the added annual household cost in the form of increased state income taxes, for the next 10 years. This type of non-voluntary payment mechanism is known to enhance a DCE's incentive compatibility and payment consequentiality (Carson and Groves, 2007; Johnston et al., 2017)[11].

The marine reserve system and blue carbon attributes both reflect potential changes in nearshore management. However, they are independent of each other in presentation, in practice, and statistically. Visually, the two attributes are presented separately, both in the background information (Figure 2: Panel A) and in each choice task (Figure 2:Panel B). The background information makes clear that blue carbon would occur across Oregon's territorial waters, with no link to the marine reserve system. In practical terms, monitoring occurs in marine reserves, but no active management is allowed. Therefore, blue carbon increases due to policy and management changes would occur outside marine reserves.

Statistically, the MR and Carbon attribute levels took the same value (i.e., both "no change" or both "50% increase") in only 12% of the choice alternatives seen by respondents. In 38% of the alternatives the MR and Carbon attribute levels had inconsistent signs (i.e., "50% decrease" for MR with "no change" or an increase in Carbon), with the remaining alternatives having the same sign but differences in magnitude across levels for the two attributes. The highly significant attribute coefficients (results below) indicate that respondents accurately understood the attribute levels and thus the independence of these two attributes.

An example choice situation is presented in Figure 2: Panel B, prefaced by an accompanying information blurb that lends context to the ensuing choice situations (Figure 2: Panel A).

[INSERT FIGURE 2 - PANEL A/PANEL B HERE]

In this example, Option 1 is associated with a 50% decrease in the size of Oregon's marine reserves, 200 new jobs gained, no change in the system's blue carbon capacity, and an associated increase in annual household cost of $400. In contrast, Option 2 is associated with a 50% increase in the size of Oregon's marine reserves, no change in jobs, a 100% increase in the system's blue carbon capacity, and an associated increase in

[11]Recently, Cunha-e-Sá et al. (2025) find that respondents were more willing to contribute voluntarily to a fund protecting mainland Portugal's vulnerable coastal region when a crowdfunding campaign was coupled with a provision-point mechanism and a money-back guarantee.

annual household cost of $100. The respondent then selects either the current situation (no change), Option 1, or Option 2, and indicates on a sliding scale from zero to 100% the certainty that the response reflects his or her preferences across the three alternatives.

Our DCE was designed to control for a host of factors affecting the accuracy, or incentive compatibility, of participant responses. These include (1) controls for both policy and payment consequentiality, (2) controls for potential biases associated with the survey's hypothetical construct, (3) availability of information necessary for respondents to understand exactly what they are being asked to value (e.g., Panel A of Figure 2), and (4) controls for both survey and attribute non-attentiveness. The survey also includes controls for response uncertainty (e.g., bottom of Panel B of Figure 2), which enables us to perform a re-coding of the data to obtain WTP estimates in the context of respondent uncertainty (c.f., Lundhede et al., 2009, Champ et al., 1997, and Li and Ando, 2023).

To address policy consequentiality, we included the following statement at the outset of the survey[12],

> *"Your responses to this survey will help coastal area managers and local policy makers better understand public opinions about 1) the state's marine reserves and 2) ocean management options that could help capture and store carbon dioxide."*

Similar to Johnston et al. (2024), payment consequentiality is controlled for organically by specifying the payment vehicle as a binding increase in annual taxes required to implement a non-status-quo alternative chosen by a respondent – increases in annual taxes being something respondents have prior experience with for payment of other local public goods. Such consequentiality also helps mitigate potential hypothetical bias (c.f., Johnston et al., 2024; Nguyen et al., 2023; Aanesen et al., 2023).

Following Penn and Hu (2019), Aadland and Caplan (2006), and Li and Ando (2023), we included the following cheap-talk reminder statement prior to the four choice situations faced by each participant to mitigate potential hypothetical bias,

> *"In the following questions, Options 1 and 2 are computer-generated. The direction and amount of change represented by the two options vary from person to person and choice to choice. Some options may not seem logical, but please treat them as realistic. Take time to consider the options in each choice, so that your responses accurately reflect your preferences. Your responses will provide important insight into the priorities of Oregon's residents. Also, please remember that if you choose Option 1 or Option 2 (and thus hypothetically choose to spend money on that option), you may have less money to spend*

[12]See Meles et al. (2023) for a nice literature review on consequentiality in choice experiments, as well as Johnston et al. (2024) and Gaur et al. (2023).

on other things you would normally purchase with your household budget. Options 1 and 2 may cost more than the current, or status quo, situation choice. The added cost to your household would occur each year for the next 10 years. With this in mind, please answer the following questions as if you really would face the consequences of your choices, which would include paying the added cost if you select Option 1 or Option 2."

4.2 Data

As described briefly in Section 4.1, a multi-step process was followed in developing and implementing the survey, which followed stated preference guidelines in Johnston et al. (2017), such as using a non-voluntary payment mechanism. Three focus groups were conducted with 10 participants each, with feedback used to modify survey content. For example, focus group participants were generally unfamiliar with the term "carbon sequestration," so it was replaced by "capture and store carbon." The survey was then pre-tested with 95 respondents, which led to additional content modification.

The final survey was completed online by two sets of respondents using the Qualtrics software platform, with attention checks used to maintain response quality. The first set of 373 respondents was from an online panel of adult Oregon residents recruited by Qualtrics and its partners. The second set of 68 respondents was recruited via postal invitations using addresses purchased from a commercial firm, based on registered drivers in Oregon. This multi-step process occurred between June 2023 and September 2024.

Respondents were excluded if they (1) protested the choice task, or did not respond to a protest survey item[13] , or (2) completed fewer than half of the full list of indicators for one or more of the included attitudinal characteristics – individual, social and environmental values. After these exclusions, the analyzed data were from 371 respondents, combined across the two sets, and a total of 1,484 choices. The data were weighted by gender to reflect population values in Oregon.

As indicated in Table 2, the U.S. census estimates for several of our sociodemographic variables fall within the 95% confidence intervals associated with their corresponding sample means. For example, the census estimate for middle-aged Oregonians (*midage*) of 40.6% falls within the 95% confidence interval for our sample mean of 40.4%. The census estimates for percentage of respondents who are Black, Asian, Pacific, or Other similarly fall within their respective 95% confidence intervals, as do the estimates for respondents with some college (*college*) or a graduate degree (*graddeg*), and those reporting high-middle incomes (*himidinc*). It is also interesting to note that the census estimate for

[13] After completing the four choice tasks, respondents indicated on a scale from 1 = strongly disagree to 7 = strongly agree the extent to which they agreed with the statement "I do not believe the government should fund marine reserves with our taxes." Respondents with missing values or with responses of 6 or 7 to this statement were excluded from the analysis.

employed individuals of 60% falls within the aggregate confidence interval for full- and part-time respondents (*workft* and *workpt*, respectively).

[INSERT TABLE 2 HERE]

Further, as Table 2 indicates, while census estimates for *senior* and *bachdeg* come close to falling within their associated confidence intervals, those for *gender, white, latino, highinc, midinc*, and *urban* do not. In particular, our sample includes higher percentages of males, whites, and middle-income earners than the general Oregon population, and lower percentages of Latinos, high-income earners, and urban households. Our empirical results should therefore be interpreted with these features of the data in mind. The data used for analysis were weighted by gender to better reflect its population distribution. Summary statistics for each of the climate urgency, relational, and recreational variables and percentage comparisons are included in Appendix C.

4.3 Estimation Strategy

As pointed out by Hensher et al. (2015), discrete choice models involve the simultaneous estimation of a number of different equations emanating from a standard random utility model (RUM) of an individual's preferences (McFadden, 1972). These equations themselves do not predict the observed choice outcomes. Rather, the equations predict latent utility for each alternative represented in the RUM, which are then subsequently used to predict the associated choice outcomes. By estimating several different choice models - both in utility- and WTP-space - we obtain estimates of consumer preferences for each attribute of the conservation scenarios, and determine which model best estimates choice behavior.

4.3.1 Econometric Models

Following Revelt and Train's (1998), Hess and Train's (2017) and Hensher et al.'s (2015) descriptions of the random-parameters (RPL), or mixed-logit (ML), version of the RUM, let person n's utility derived from alternative j in choice situation t be denoted as,

$$u_{njt} = \boldsymbol{\beta}_{\boldsymbol{n}}' \boldsymbol{X}_{njt} + \varepsilon_{njt}, \qquad n = 1, \ldots, N,\ j = 1, \ldots, J, \text{ and } t = 1, \ldots, T. \tag{1}$$

where $\boldsymbol{X}_{njt}$ is a vector of given attribute levels, $\boldsymbol{\beta}_{\boldsymbol{n}}$ is a corresponding, individual-specific vector of marginal utility coefficients associated with each of k attributes comprising the environmental good in question, and ε_{njt} is an idiosyncratic error term representing the unobserved component of utility, generally assumed iid extreme value type I. Vector $\boldsymbol{X}_{njt}$ not only includes attribute levels, but also alternative specific constants (ASCs) (which can vary across individuals) and individual-specific demographic and attitudinal characteristics. Vector $\boldsymbol{\beta}_{\boldsymbol{n}}$ is typically modeled as,

$$\boldsymbol{\beta}_n = \bar{\boldsymbol{\beta}}_n + \boldsymbol{\eta}_n \boldsymbol{\nu}_n, \qquad n = 1, \ldots, N. \tag{2}$$

where $\bar{\boldsymbol{\beta}}_n$ represents the vector of the means of the distributions of the k marginal utility coefficients exhibited by the sampled population, $\boldsymbol{\eta}_n$ represents the vector of spreads, or standard deviations, of preferences for the k attributes around $\bar{\boldsymbol{\beta}}_n$, and $\boldsymbol{\nu}_n$ denotes random draws taken from pre-specified distributions for each respondent and attribute k (Hensher et al., 2015).

Under these assumptions, person *n's* probability of choosing alternative j in choice situation t conditional on $\boldsymbol{\beta}_n$ can be denoted as,

$$L_{njt} = \frac{e^{\boldsymbol{\beta}_n^\top \boldsymbol{x}_{njt}}}{\sum_{i=1}^{J} e^{\boldsymbol{\beta}_n^\top \boldsymbol{x}_{nit}}}, \qquad n = 1, \ldots, N,\ i \neq j,\ j = 1, \ldots, J,\ t = 1, \ldots, T. \tag{3}$$

where e is Euler's number. Denoting the multivariate, continuous, cumulative distribution function (CDF) of $\boldsymbol{\beta}_n$ as $F(\boldsymbol{\beta}_n \mid \boldsymbol{\theta})$, where the elements of $\boldsymbol{\beta}_n$ may be correlated with each other, and $\boldsymbol{\theta}$ represents a vector of parameters (common across individuals) defining $F(\cdot)$, the probability of individual n's choice sequence can be written as

$$P_n = \int L_{njt}\ f(\boldsymbol{\beta}_n \mid \boldsymbol{\theta})\, d\boldsymbol{\beta}_n, \qquad n = 1, \ldots, N, \tag{4}$$

where $f(\cdot)$ is the density function associated with $F(\cdot)$. Given a specification for $F(\cdot)$, the researcher estimates its parameters, $\boldsymbol{\theta}$, using simulation (since there is no closed-form solution to the integral in equation (4)) [14].

As pointed out by Hess and Train (2017), correlations among the elements of $\boldsymbol{\beta}_n$ for any given individual n may arise for several different reasons, the most direct being that an individual's observed preference for one attribute is related to his preferences for other attributes. For future reference, we refer to this phenomenon as "coefficient correlation", captured by the off-diagonal elements of the covariance matrix of $\boldsymbol{\beta}_n$.

In addition, unobserved factors could be correlated with an individual's choices across different choice situations, which in turn induces correlation among observed preferences for the different attributes. This phenomenon is known as "scale heterogeneity" because the scale of utility (i.e., the magnitude of all utility coefficients in $\boldsymbol{\beta}_n$) differs across individuals.

Accounting for scale heterogeneity in the context of the RPL model results in a Generalized Mixed Logit model (GMXL), with equation (1) being rewritten as,

$$u_{njt} \ = \ \beta_n' x_{njt} \ + \ \frac{1}{\sigma_n}\, \varepsilon_{njt}, \qquad n = 1, \ldots, N;\ j = 1, \ldots, J;\ t = 1, \ldots, T, \tag{5}$$

[14] The simulation is carried out via a quasi-Monte Carlo approach, where values are sampled from the support of $f\,(\beta_n|\theta)$ through inverting Halton sequences. Taking the average of evaluations for Halton sequences of length R results in $\widetilde{P}_n = [1/R] \sum_{r=1}^{R} L_{njt}$ (Hensher and Green, 2003).

where scale parameter σ_n is inversely proportional to the standard deviation of ε_{njt}, generally assumed to be log-normally distributed with normalized mean equal to one for estimation purposes ($E[\sigma_n] = 1$)[15]. In turn, because u_{njt} is unit-less, equation (5) can be rewritten as,

$$u_{njt} \;=\; (\sigma_n \boldsymbol{\beta}_n)' \boldsymbol{X}_{njt} \;+\; \varepsilon_{njt}, \qquad n = 1, \ldots, N;\; j = 1, \ldots, J;\; t = 1, \ldots, T, \tag{6}$$

From equation (6) we see that, if it indeed exists, scale heterogeneity (represented by parameter σ_n), imposes correlation among the full set of an individual's utility coefficients (represented by $\boldsymbol{\beta}_n$) – correlation that differs across individuals. The larger is σ_n, the greater is the correlation among utility coefficients $\boldsymbol{\beta}_n$. Models designed solely for scale heterogeneity (where the elements of $\boldsymbol{\beta}_n$ are assumed uncorrelated), are thus restricted forms of RPL containing a single correlation parameter (represented by σ_n). In this case, parameter σ_n captures the combination of whatever sources of correlation exist in the data among the utility coefficients (thus, contrary to how it is designed, σ_n cannot be interpreted as solely representing scale heterogeneity)[16].

In addition, Hess and Train (2017) point out that σ_n and coefficient correlation are generally not separately identified from standard choice data. Thus, an RPL specification (e.g., equation (1)) that allows for full coefficient correlation induced by behavioral sources, also accounts for any correlation induced by scale heterogeneity. The authors argue that the RPL specification is the most general form capturing both coefficient correlation and scale heterogeneity, and distinguishing the two requires additional structure or information.

By contrast, Fiebig et al. (2010) and Hensher and Greene (2011) propose a more generalized version of the GML model, with β_n in equation (6) defined as

$$\boldsymbol{\beta}_n \;=\; \boldsymbol{\theta}_n \bar{\boldsymbol{\beta}}_n \;+\; \Big[\gamma + \boldsymbol{\theta}_n (1-\gamma)\Big]\, \boldsymbol{\eta}_n \boldsymbol{\nu}_n, \qquad n = 1, \ldots, N. \tag{7}$$

Here $0 \leq \gamma \leq 1$ and $\boldsymbol{\theta}_n = \exp\Big(-\frac{\tau^2}{2} + \tau\, \omega_n\Big)$ with $\omega_n \sim \mathcal{N}(0,1)$.

Hence, the GMLX model can be viewed as nesting both the RPL and the multinomial logit models—when $\tau = 0$ the GMXL specification results in the RPL model, and when γ=0, the GMLX results in a scaled version of a multinomial logit model.

In addition to RPL models with and without coefficient correlations and GMLX models, we report results associated with three additional specifications: (1) RPL with error components, (2) models in WTP space and (3) equality-constrained latent class models.

[15] Comparing equation (6) with (1), we see that the ML model effectively assumes σ_n=1.

[16] As Hess and Train (2017) point out, σ_n could conceivably be found to be statistically significant when in fact scale heterogeneity does not exist. In this case, the researcher can conclude that "the hypothesis cannot be rejected that the correlation pattern among coefficients takes the form that would arise if scale heterogeneity were the only source of correlation (page 4)."

The standard RPL model can be extended to accommodate additional structure in the unobserved stochastic component of the utility, particularly to account for correlation across alternatives that is not captured by the observed attributes (Train, 2009). This leads to the Random Parameter Logit model with Error Components (RPL-EC), which adds an alternative-specific unobserved utility component. The utility function under this specification is expressed as:

$$U_{njt} = \boldsymbol{\beta}_n^{'} \boldsymbol{X}_{njt} + 1_j(\mu_{njt}) + \varepsilon_{njt}, \qquad n = 1, \ldots, N;\ j = 1, \ldots, J;\ t = 1, \ldots, T, \tag{8}$$

where $\mu_{njt} \sim N(0, \sigma_\mu^2)$ is the normally distributed error component shared across a subset of alternatives capturing unobserved similarities (e.g. shared "policy package" effects) and $1_j(\cdot)$ is an indicator function equal to 1 if the alternative j is assumed to share the common error component and 0 otherwise[17]. This structure introduces additional correlation in utility across alternatives and allows for more realistic substitution patterns - alternatives with shared error components compete less strongly against each other (i.e., they are perceived as closer substitutes).

In both RPL and RPL-EC models estimated in utility space, the implied willingness to pay (WTP) for individual n for a non-monetary attribute k is computed as the ratio of the marginal utility of the attribute to the marginal utility of the cost attribute:

$$\text{WTP}_n^k \ = \ -\frac{\beta_n^k}{\beta_n^p}, \quad k = 1, \ldots, K-1 \tag{9}$$

where β_n^k is the individual-specific coefficient for attribute k, and β_n^p is the individual-specific coefficient for the cost (price) attribute, p. Because both the numerator and denominator in (9) are random variables, the resulting WTP distribution is a ratio of random variables, which can be highly skewed and may result in undefined or infinite statistical moments[18] (Train and Weeks, 2005; Daly et al., 2012; Thiene and Scarpa, 2009).

To overcome these limitations, Scarpa et al. (2008) and Hensher and Greene (2011) propose estimating the model directly in WTP-space rather than utility space. This approach normalizes utility in monetary units constraining the cost coefficient to equal -1, thereby expressing parameters directly as WTP measures. Substituting equation (9)

[17]We follow Scarpa et al. (2005) and assume that because the status quo alternative is actually experienced by respondents and repeated in each choice task, while alternatives 1 and 2 are hypothetical and change in each choice task, it is more likely that the unobserved utility of alternatives 1 and 2 is distributed with the same variance and is more correlated than either of these two alternatives with the status quo. Thus, we specify the common error component - where the indicator function equals 1 when the alternatives are 1 or 2, zero otherwise.

[18]In this research the Delta Method with 2,000 draws is implemented to determine the significance of mean WTP estimates derived from utility space models along with corresponding confidence intervals (Ellis et al, 2019).

into (8), the WTP-space version of the RPL-EC model is given by:

$$U_{njt} = \beta_n^p \Big(- \text{Price}_n \; + \; \mathbf{WTP}_n^{'} x_{njt} \; + \; 1_j(\mu_{njt}) \; + \; \frac{\varepsilon_{njt}^p}{\beta_n^p} \Big) \tag{10}$$

where $\mathbf{WTP}_n$ is the vector of individual-specific willingness-to-pay values for each attribute, and utility is scaled by the individual-specific cost coefficient β_n^p. As Hess and Train (2017) point out, scale heterogeneity in this model is represented by the inverse of β_n^p.

Lastly, we consider the equality constrained latent class (ECLC) model, which provides an alternative approach to modeling heterogeneity among survey respondents. Here, it is assumed that the population consists of a finite number, C, of groups of individuals. Individuals within a given group are considered homogeneous. However, the groups themselves differ from one another[19].

In its standard form, the ECLC model assumes that individuals are implicitly sorted into a discrete set of C classes. However, which class contains which individual is unknown by the analyst, hence the moniker "latent class" (Hensher et al., 2015; Wallmo and Edwards, 2008). To reflect this type of latent heterogeneity, we can rewrite equation (3), which now reflects person n's probability of choosing alternative j in choice situation t conditional on being in class c, as,

$$L_{njt}|_c = \frac{e^{\boldsymbol{\beta}_c' x_{njt}}}{\sum_{i=1}^{J} e^{\boldsymbol{\beta}_c' x_{nit}}}, \; n = 1, \ldots N, \; i, j = 1, \ldots J, \; t = 1, \ldots T, \text{ and } c = 1, \ldots C, \tag{11}$$

where coefficient vector $\boldsymbol{\beta}_c$ is class- rather than individual-specific (in a sense, discretely rather than continuously distributed). In the equality-constrained formulation of this model, the $\boldsymbol{\beta}_c$ vector elements are constrained to be the same across classes for attended attributes. This leads to a revision of equation (4) – the probability of observing individual n's choice sequence – as,

$$P_{n|c} = \int_{t=1}^{T} \left[L_{njt|c} \right] f_c(\boldsymbol{\beta}_c \mid \boldsymbol{\theta}_c) \, d\boldsymbol{\beta}_c, \qquad n = 1, \ldots, N, \; c = 1, \ldots, C. \tag{12}$$

where $\boldsymbol{\theta}_c$ represents a vector of parameters (common across individuals in class c) defining class-specific density function, f_c. Similar to the probability of her choice behavior given her assignment to class c, the prior probability of individual n being assigned to class c can be written as,

$$H_{nc} = \frac{e^{\gamma_c' \boldsymbol{z}_n}}{\sum_{c=1}^{C} e^{\gamma_c' \boldsymbol{z}_n}}, \; n = 1, \ldots N, \; c = 1, \ldots C, \tag{13}$$

where $\boldsymbol{z}_n$ again represents a vector of respondent n's personal characteristics, and vector

[19] More recently, models with random parameters imposed within each class have also been estimated, c.f., Czine et al. (2024).

γ_c is a corresponding vector of estimable coefficients (Hensher et al., 2015; Wallmo and Edwards, 2008; Greene and Hensher, 2003).

The ECLC model lends itself to testing for the ex post implied, or inferred, presence of attribute ANA, particularly ANA of the cost attribute in any given alternative (c.f., Li and Ando, 2023), as well as testing for the presence of common-metric attribute aggregation (ACMA) due to a model's inclusion of separate attributes that are nevertheless measured in a common metric. Scarpa et al. (2009) point out that analyzing the observed response pattern among respondents using a statistical model with degenerate distributions of taste intensities at zero identifies non-attendant behavior. We follow Scarpa et al. (2009), Petrolia and Hwang (2020) and Li and Ando (2023) in constraining the values of non-zero coefficients to be the same across all classes ("equality constrained"), so that only attendance and non-attendance, but not taste heterogeneity across respondents, are captured by the model.

As explained in Section 6, our analysis captures both this type of ex post inferred ANA, as well as ex post stated ANA, since we included the following statement at the conclusion of the DCE,

> *"I did not consider selecting the current situation (no change) option when making my four choices,"*

to which the respondent selected from 1 (strongly disagree) to 7 (strongly agree).

5 Empirical Results

In this section, separate sets of results are presented for linear and non-linear specifications of the DCE models described in Section 5 – models estimating the effects of different attribute levels (ref. Table 1) on the typical respondent's preferences. The non-linear models provide estimates of separate preferences for increases and decreases in the attribute levels, revealing the extent of asymmetries between the two. The extent to which the magnitude of WTP for a decrease in an attribute level (i.e., negative WTP) exceeds that for an increase in the level (positive WTP), indicates the degree of loss aversion associated with that attribute. The non-linear models also provide convenient points of departure for scenario analysis. For the linear specifications, we also derive estimates of marginal WTP in the presence of respondent uncertainty, as described in Section 4.1.

We test for preference heterogeneity in two ways. First, we estimate random parameter specifications that allow for individual-specific variation in coefficient and WTP estimates. Second, to measure potential heterogeneity driven by respondents' recreation behaviors, we compare results across subsamples of respondents who recreate often (particularly with respect to fishing) versus those who do not. (Malone and Lusk, 2018). We conclude

with a discussion of the estimation results that test for potential ANA in its myriad forms. [20]

5.1 Linear Models

Tables 3 and 4 present results for four separate specifications – two in preference space and two in WTP space[21]. The preference-based models are random parameters logit (RPL) and RPL with one error correction term (RPL-EC). The WTP-space models are RPL-EC estimated in WTP space (RPL-EC-WTP) and generalized mixed logit in WTP space (GMLX-WTP) models. The RPL model is meant to serve as a baseline of comparison with the other models. The RPL-EC model corresponds to Chen et al.'s (2023) Case 1 structure, where the two policy-change alternatives in our DCE (Alternatives 1 and 2) share the same EC term, implying that the utility levels of these alternatives are correlated but independent from that of the status quo. This model was chosen based upon its superior goodness-of-fit measures viz-a-viz the other models estimated in preference space.

Following Hess and Train's (2017) general recommendations, we ran each model assuming correlated coefficients. However, the overall model fit for the GMXL-WTP model improved when relaxing this assumption, and the standard deviation estimates for the coefficients in the RPL-EC-WTP model became statistically significant. Hence results for these two models are reported for the uncorrelated-coefficient specifications[22]. Each model assumes normal distributions for the non-cost coefficients (including the Alternative Specific Constant (ASC) for the status quo alternative) and a triangular distribution for the cost coefficient[23].

In Table 3, we see that, as expected, the coefficient estimates for each of the non-cost attributes are positive and statistically significant in each model, indicating that increases in the attributes' levels are associated with utility increases for the typical respondent (recall that the coefficient estimates for the RPL-EC-WTP and GMXL-WTP models are the marginal WTP estimates themselves). The SQ coefficient is negative and statistically significant, indicating that, all else equal, respondents value either of

[20] In Lindberg et al. (2026), We investigate preference heterogeneity in the context of a hybrid-choice model, where preferences are moderated by perceived climate urgency, which in turn is predicted by relational values and anthropocentric environmental attitudes and personal values of self-transcendence and conservatism.

[21] We ran several other specifications of the model. These are the best performing of the specifications. The multinomial logit model (MNL) was also estimated. However, the Hausman and McFadden (1984) specification test of the independence from irrelevant alternatives (IIA) resulted in a χ^2 statistic of 17.95 (pr = 0.0013), indicating rejection of the IIA assumption and thus the MNL model's results.

[22] Statistically significant negative cross-correlations in the RPL and RPL-EC models occur between the jobs and reserves, carbon and asc-sq, and carbon and jobs coefficients, while a positive cross-correlation occurs between the cost and reserves coefficients.

[23] Models assuming the log-normal distribution for the cost coefficient did not converge. Hence, we follow Hensher et al. (2015) in assuming a one-sided triangular distribution.

the two alternatives over the status quo. The cost estimate is, as expected, negative and statistically significant in the preference-space models. Further, variances associated with the coefficient estimates are, with few exceptions, statistically significant, indicating the statistical relevance of preference heterogeneity across individuals.

The χ^2 statistics for each model in Table 3, as they are for all results reported in this paper, are statistically significant at the 1% level, indicating that we can reject the hypothesis of jointly zero coefficient estimates. The corrected AIC, BIC, and HIC (Hannan Quinn) measures (corrected for sample size) indicate superior fits for the RPL, RPL-EC, and RPL-EC-WTP models, as do the McFadden's Pseudo R2 measures and log-likelihood values. Interestingly, the cross-alternative equality constraint on the estimated standard deviations in the RPL-EC model is statistically insignificant, while in the RPL-EC-WTP model it is statistically significant at the 1% level. The scale heterogeneity statistic τ for the GMXL-WTP model is statistically insignificant, indicating that the unobservable error variance does not vary statistically across respondents. Consequently, the data provide little evidence that some individuals exhibit systematically higher "noise behavior" than others (i.e., that some people are more random than others when making choices). Therefore, this idiosyncratic component of utility can be treated as homoscedastic and the heterogeneity in the estimated coefficients and WTP mainly emanate from differences in preferences, not from differences in scale or error variance.

[INSERT TABLE 3 HERE]

Corresponding marginal WTP estimates (Table 4) in utility space are calculated using the Delta Method utilizing 2,000 random draws. We see some uniformity across models regarding the magnitudes of the estimates (all estimates are statistically significant at the 1% level). The annual premiums for a percentage increase in MR system size ($2 to $3.56), and blue carbon ($1.71 to $3.29) are higher compared to an additional unit of job creation ($0.66 to $0.98). The RPL-EC model in WTP-space reduces the MR coefficient in relation to Carbon. This indicates that not controlling for the unobserved, shared component between these two attributes results in unrealistic relative WTP estimates between Carbon and MR system size. Also, ignoring the shared component results in a higher, unrealistic avoidance of status quo. The SQ scenario is consistently rejected, and people are willing to pay to avoid it (from $110 to $238), revealing that, on average, Oregonians can achieve utility gains from changes to Oregon's nearshore spatial management. Breeze et al. (2015) interpret this coefficient estimate as an existence value (i.e., the innate value respondents attach to knowing that a policy is in place).

[INSERT TABLE 4 HERE]

Table H in Appendix C contains WTP estimates adjusted for response uncertainty, as mentioned in Section 4.1 (a re-coding of the data following Lundhede et al., 2009, Champ et al., 1997, and Li and Ando, 2023). In our sample, 113 respondents (roughly 30% of our sample) self-identified a certainty level of 50% or lower in at least one of

their four choice situations. These respondents were at or below the 50% threshold on a total of 250 choice situations, and of those, 151 resulted in re-coding to the status quo alternative. As shown in Table H, the WTP for the status quo falls dramatically in each model, from highly negative in the models without controlling for uncertainty (except RPL-EC in utility-space) to statistically insignificant or more negative. Interestingly, the WTP estimates for MR rises consistently in the WTP-space models and for Carbon in RPL-EC models. Except for the RPL-EC-WTP model, WTP estimates for net job creation decrease. Unlike in the dichotomous-choice models of Champ et al. (1997) and Lundhede et al. (2009), where the WTP estimates expectedly decrease, there was no expectation in our models of a similar effect. This is because the attribute levels of MR and jobs reflect both increases and decreases in values relative to the status quo.

5.2 Non-Linear Models – Loss Aversion, and Tradeoffs

Our results for the non-linear models are presented in Tables 5 and 6[24]. We consider WTP for discrete changes in the attribute levels, in turn enabling us to test for the existence of loss aversion [25] among our survey respondents and, in the case of carbon, diminishing WTP in the amount of blue carbon. In Tables 5 and 6, we see strong evidence of loss aversion. For instance, in Table 5, for the RPL-EC model, the effect on utility of a 50% decrease in the size of the MR (MR(-50%)) of -23.45 is noticeably larger in magnitude than the effect of a 50% increase in size (MR(+50%)) of 8.28, translating to respective annual WTP values of -$222 and slightly higher than $78. Even more pronounced is the relationship between jobs(-200) and jobs(+200), where the magnitude of loss aversion is almost $150, going from -$149 for the decrease in jobs to a statistically insignificant $46.

[INSERT TABLE 5 HERE]

[INSERT TABLE 6 HERE]

In both the RPL-EC and RPL-EC-WTP models we find evidence of a U-shaped relationship between the amount of blue carbon and corresponding WTP, with the latter model showing a steep rise in WTP from carbon(+50%) to carbon(+100%) of over $100. Only the RPL model finds a diminishing relationship, and then with only $20 separating the highest from lowest estimates. Comparing the results in Table 5 with those in Table 3 for the linear specifications, we see that the non-linear specifications generally do not fit the data as well. Further, we find less statistical significance for the standard deviations of the random coefficients across both types of models.

Results for our tradeoff analysis are presented in Table 7. Here, we interact each

[24] Results for the GMXL-WTP model have been dropped due to its poor overall fit and lack of any statistically significant coefficient estimates.

[25] Loss aversion is the tendency to weigh potential losses more heavily than equivalent gains, producing a kinked value function. This creates a change in risk aversion around the reference point: risk-averse for prospects framed as gains, risk-seeking for prospects framed as losses (Kahneman and Tversky,1979).

attribute with the other, focusing on the signs and statistical significance of the interaction terms (magnitudes of these estimates are generally low in value, indicating relatively minor tradeoffs compared with the magnitudes of the independent attribute estimates themselves). These models are estimated solely in preference space (RPL and RPL-EC specifications). They assume uncorrelated parameters to isolate the tradeoffs independently of correlations estimated within the model. For similar reasons, the interaction terms are estimated as fixed rather than random coefficients.

[INSERT TABLE 7 HERE]

From Table 7 we see that, across both specifications, MR system size and blue carbon (MR x Carbon) and jobs and blue carbon (Jobs x Carbon) exhibit statistically significant negative tradeoffs, i.e., respondents are willing to tradeoff a larger MR for a smaller level of blue carbon (and vice-versa), as well as a larger number of jobs for a smaller level of blue carbon (again, vice-versa). To the contrary, MR and Jobs (MR x Jobs) and blue carbon and cost (Carbon x Cost) exhibit positive tradeoffs, implying that the value of the MR increases with increases in the number of jobs - indicating complementarity - and the value of blue carbon increases with increases in the costs associated with the policy change.

This latter result suggests a price-signaling effect, whereby a higher program cost signals higher value in blue carbon to the typical respondent. In other words, the negative impact of cost is moderated when blue carbon increases. This result has important policy implications as well. For example, it suggests that the public supports investment in blue carbon even when it costs more – people are not choosing the cheapest option. Our data suggests that Oregonians are willing to pay more when additional carbon is sequestered. Consequently, investment in blue carbon projects – and related funding mechanisms such as conservation fees or taxes – may be likely to receive public support.

5.3 Scenario Analysis

Marginal WTP estimates for three separate policy scenarios were derived using attribute coefficients from the linear models presented in section 5.1. For models estimated in utility space, the WTP values are calculated by multiplying the estimated attribute coefficients by the respective levels defined in each scenario and dividing the result by the corresponding cost coefficient. For WTP-space specifications, the scenario-specific WTP values are obtained by directly multiplying the estimated WTP coefficients by the marginal levels of the respective policy attributes. The three policy scenarios are defined as follows: 1) *Optimistic* (50% increase in MR system size + 200 additional jobs + 100% increase in blue carbon); 2) *Status Quo/ASC* (no change in MR system size, employment, and blue carbon); 3) *Pessimistic* (50% reduction in MR system size + 200 fewer net jobs + 0% increase in blue carbon).

The WTP results for distinct specifications (Table 8) indicate that people on average are willing to spend an annual amount between \$464 to \$704 over ten years for the *Optimistic* scenario. Oregonian's citizens are willing to spend an annual amount ranging from \$110 to \$238 to avoid keeping the nearshore spatial management situation as it is for each of the next ten years, and \$236 to \$374 to avoid the *Pessimistic* scenario. It is interesting to note that both WTP-space models show evidence of WTP overshoot, whereby both WTP estimates for the Optimistic scenario exceed the study's highest cost level of \$500 (Glenk et al., 2024). The estimates from the preference-based models do not suffer this face-validity problem.

[**INSERT TABLE 8 HERE**]

Posterior cumulative WTP distributions for these three scenarios and additional scenarios with attributes prioritized - in the face of budget constraints - are presented in Appendix E. The distributions are based upon estimated individual WTP values derived from the RPL-EC-WTP model. They indicate that, for instance, over 50% of survey respondents are willing to pay at least \$600 per year for the *Optimistic* scenario and 80% are willing to pay a positive amount to avoid the *StatusQuo*. Also, prioritizing high blue carbon levels is key, showing that a *blue carbon-intensive scenario* even with 50% lower job levels and MR system size, respectively, provides almost the same 60% participation rate as the *Optimistic* scenario.

5.4 Preference Heterogeneity

We compare two different sets of subsamples of the data based upon the extent to which individuals recreate along the Oregon coast. We estimate RPL-EC models in WTP space assuming both correlated and uncorrelated parameter specifications (the analogous results for RPL and RPL-EC models in utility space, not accounting for correlation between parameters, are contained in Appendix A). The first pair of subsamples distinguishes individuals who had previously participated in fishing recreational activities (fishers) from those who had not (non-fishers)[26]. These two subgroups (n_1=189 and n_2=159, respectively) were defined by creating a dummy variable equal to 1 for individuals who answered that they had engaged in "Ocean, recreational fishing (including charter) and or collecting shellfish (e.g crabs or clams)" at least once in the past 5 years, equal to zero otherwise.

Conversely, the second pair of subsamples differentiates respondents who had previously engaged in any general coastal recreational activity from those who had not. The two subgroups (n_3=159 and n_4=118 respectively) are defined by creating a dummy variable equal to 1 for individuals who reported participating more than 10 times in the

[26]We focus on household WTP of residents and recreation heterogeneity; commercial fishery impacts are acknowledged in costs and job counts but not modeled as a distinct choice class

previous five years in any of the following activities: ocean-oriented sightseeing, photography, and/or wildlife viewing (birds, whales etc.), beach walking, running, shell collecting and/or rockhounding, and exploring tide pools, zero otherwise.

Because the "Fishing Recreation" and "No-Fishing Recreation" sub-samples are mutually exclusive, the pooled model is not nested within a combined segmented model, precluding extraction of observation-wise log-likelihoods. Therefore, we compare model quality using the Akaike Information Criterion (AIC) and the Bayesian Information Criterion (BIC). We observe that $\text{AIC}_{\text{subsampled}} < \text{AIC}_{\text{pooled}}$ and $\text{BIC}_{\text{subsampled}} < \text{BIC}_{\text{pooled}}$ for both models estimated with and without correlated parameters[27]. This indicates that estimating separate models provides a better fit of the data, where the gain in fit from estimating separate equations more than compensates for the additional parameters, yielding a better balance of goodness-of-fit (fitness) and parameter economy (parsimony).

To assess whether recreational fishers' willingness to pay (WTP) for each attribute stochastically dominates that of non-fishers, we first simulated 10,000 correlated and uncorrelated random draws of the WTP-space coefficients from each RPL-EC specification with assumed correlated parameters. Specifically, for each pair of subsamples (Fishing Recreation vs. No-Fishing Recreation), we supply the estimated mean vector ($\boldsymbol{\mu}$) and full variance-covariance matrix ($\boldsymbol{\Sigma}$), simulating draws from a multivariate normal distribution. We then apply two complementary one-sided tests: Poe et al.'s (2005) convolution test for independent samples, and Mann–Whitney U tests (Hollander and Wolfe, 1999) to check whether the median WTP of one model exceeds the median of the other as a non-parametric, distribution-free check, without relying on strong distributional assumptions.

The results in Table 9 confirm that the WTP coefficients of SQ in the uncorrelated and correlated model comparisons show clear evidence of significantly different WTP estimates across subsamples by both the convolution and Mann–Whitney U tests: fishers exhibit higher WTP to avoid the status quo than non-fishers, i.e., non-fishers are less inclined to reject the SQ of Oregon's nearshore spatial management than fishers. We also find evidence (according to the Mann–Whitney U tests) that non-fishers place greater value on increases in each of the remaining attributes for all specifications. This in turn indicates that non-fishers value nearshore spatial management attributes more highly, perhaps because they seek broader amenity and climate benefits, not prioritizing catch-related gains. Recreational fishers, though assigning lower WTP to these enhancements, experience greater welfare losses under current conditions, and therefore demand higher compensation to accept the SQ. These patterns are consistent with recreational fishers perceiving the SQ as risky because it may permit future expansion of commercial fishing

[27]For the RPL-EC subsampled models assuming correlated parameters: $AIC_{subsampled} < AIC_{pooled}$ = (1392.4+ 1073.9) < 2626.6 and $BIC_{subsampled} < BIC_{pooled}$ = (1489.6 + 1167.5) < 2737.9. For the RPL-EC subsampled models assuming uncorrelated parameters: $AIC_{subsampled} < AIC_{pooled}$ = (1404 + 1085.9) < 2626.6 and $BIC_{subsampled} < BIC_{pooled}$ = (1454.9 + 1135) < 2737.9.

locally.

[INSERT table 9 HERE]

The Mann–Whitney U tests evince shifts in medians across entire distributions, using ranks of the values rather than their numeric magnitudes. Hence, the test is sensitive to differences in central tendencies—such as median shifts. When the sample size is large, even modest differences in rank ordering can result in statistically different median differences, leading to more frequent rejection of equal WTP values across subsamples. The convolution test, on the other hand, is designed to detect first-order stochastic dominance, a stricter criterion that requires that one distribution yield consistently higher values than the other across the entire range of values. [28]

Table 10 shows that estimating separate models distinguished by general recreation frequency also results in a better fit of the data than a pooled model[29]. The marginal WTP estimates (in magnitude) for SQ and blue carbon are larger for the subsample of individuals who regularly engage in general recreational activities on the Oregon coast than for those who do not. Conversely, marginal WTP estimates for respondents who do not regularly engage in general recreation are larger than for those who do.

At the same time, the number of jobs created presents inconsistent patterns across the uncorrelated and correlated specifications: higher WTP estimates are obtained for those who regularly engage in recreation when accounting for correlated parameters and lower estimates with uncorrelated parameters. All WTP differences are statistically significant as confirmed by the Mann-Whitney U tests.[30]

In short, general recreationists express lower negative WTP (in magnitude) for the status quo, and larger WTP enhancing blue carbon, suggesting stronger preferences for climate and ecosystem stability. In contrast, non-recreational users express relatively greater WTP for increasing the MR system size, suggesting that recreational users may perceive that the activities they engage in could be restricted or diminished by an expansion of the MR system. The mixed pattern observed in job creation suggests that preferences for local economic benefits may not be uniformly aligned with recreation frequency. Together, these findings highlight differentiated priorities among user groups that can inform more targeted and effective coastal policy design.

[28]Specifically, the test determines whether the proportion of pairwise differences $D_{ij} = x_i - y_j$ across subsamples—where x_i is the WTP draw from subsample i and y_j is the corresponding WTP draw from subsample j—exceeds zero in more than 95% of all possible cross-group comparisons.

[29]For the RPL-EC model assuming correlated parameters: $AIC_{subsampled} < AIC_{pooled} = (1647.8 + 886.4) < 2626.6$ and $BIC_{subsampled} < BIC_{pooled} = (1749.1 + 973.7) < 2737.9$. For the RPL-EC model assuming uncorrelated parameters: $AIC_{subsampled} < AIC_{pooled} = (1668.2 + 834.9) < 2626.6$ and $BIC_{subsampled} < BIC_{pooled} = (1721.3 + 880.6) < 2737.9$.

[30]This result holds even when the two distributions overlap considerably, that is, when individual WTP values from both subsamples are not fully distinct, e.g., some values from the lower-median group still exceed those from the higher-median group. To the contrary, the statistical non-significance found the convolution tests for each WTP attribute suggests that the two subgroups' distributions overlap substantially, that is, the values are not clearly separated. The degree of overlap prevents convolution tests from finding a strict dominance, even if one group generally tends to exhibit higher WTP estimates.

[INSERT table 10 HERE]

A fuller analysis of fishing-status interactions and comparisons of recreationists and non-recreationists is provided in Appendix H.

5.5 Attribute Non-Attendance (ANA)

Similar to testing for preference heterogeneity, we test for ANA using two approaches. The first approach tests for ex post stated ANA based upon responses to the statement "I did not consider selecting the current situation (no change) option when making my four choices," as described in Section 5. For this analysis, we created a dummy variable equaling one if the respondent agreed or strongly agreed with the statement (i.e., chose 6 or 7 on the Likert scale), zero otherwise. Twenty-seven percent of sample participants agreed or strongly agreed with the statement. We then interacted this variable with each respective attribute to assess the extent to which the attribute values are affected by this measure of ANA. WTP results are presented in Table 11.

[INSERT table 11 HERE]

We see from the table that individuals who (stated that they) ignored, i.e., did not attend, the status quo when making their choices expressed higher WTP for blue carbon than those who were attentive to the status quo. The former group's WTP premium is roughly $4.70 more for an additional percent increase in blue carbon. Interestingly, those ignoring the status quo expressed a WTP discount of roughly $1.70 for an additional percent increase in the size of the MRS (potentially an artifact of the reserve attribute including a negative value of -50%). There was no statistically significant effect of this form of ANA on WTP for additional jobs. The second method follows the ex post implied ANA approach of Hensher et al. (2015), Scarpa et al. (2009), Campbell et al. (2011), and McNair et al. (2012) in estimating an ECLC model described in Section 5. Results from a model with full attribute attendance (FAA) are compared with those obtained from a model accounting for potential ANA associated with the status quo (SQ) and cost attributes. This method therefore controls for a wider range of potential ANA than the former ex post stated approach. Table 13 presents WTP results for the model with potential ANA in SQ and $Cost$. The model assumes three classes (to align with its FAA and two ANA specifications) and fixed coefficients (Hensher et al., 2015). The full set of results are contained in Appendix E, Tables J and K.

[INSERT table 12 HERE]

The most relevant comparison of the results in Table 12 is with those reported in Table 4. We see that the model controlling for potential ANA reports a larger WTP to avoid the status quo, but lower WTP estimates for the remaining attributes. The larger WTP to avoid the status quo makes intuitive sense, since both the status quo and cost are unattended in the ANA model. Similar to previous findings, the smaller WTP

estimates for the remaining attributes are likely artifacts of negative values included in the attributes' respective ranges of levels.

We conducted a Vuong (1989) test of the two (non-nested) models represented by the results in Tables 4 and 13. Our Vuong statistic of 16.56 is statistically significant at the 1% level, indicating that the results in Table 4 better explain the data generating process than those in Table 12. Comparison of the results in Tables J and K suggest further caution in interpreting the validity of ANA results. Although the latent class probabilities are generally similar (and each statistically significant at the 1% levels for both models), and the associated coefficient estimates are likewise similar across the two (FAA and ANA) models, the goodness-of-fit measures all suggest that the FAA model in Table L fits the data better than the ANA model in Table 12. We also conducted a Likelihood Ratio (LR) test comparing the two models. The LR statistic (χ^2) of 193.16 is statistically significant at the 1% level, suggesting that the restrictions of the ANA model do not improve the overall fit of the data. Hence, we conclude that the FAA model is superior in terms of overall fit of the data.

6 Discussion

This study has employed a host of different modeling approaches designed to estimate Oregonians' preferences for the size of the state's marine reserve system, the num-ber of coastal jobs, and change in blue carbon resources. The models are estimated in both preference- and WTP space, with and without error component, correlation of preferences, and potential attribute non-attendance in the model's different attributes, and in linear and non-linear frameworks, with the primary goal being to extend empirical results from previous studies of Oregon's coastal environment, particularly with respect to the valuation of marine reserve system size and blue carbon.

Our findings indicate that, all else equal, Oregonians exhibit monotonic preferences for increasing the size of the marine reserve system, coastal employment, and blue carbon. These results align with prior research confirming high public acceptability for marine reserve enhancements (Abate et al, 2020; Ansen, 2015, Mazzoco et al, 2022; Nguyen et al, 2023), as well as blue carbon (Riegel et al., 2023; Tyllianakis, 2022). We find that annual WTP for (1) expanding the size of the marine reserve system ranges from $2.05 to 3.57$ per percent increase, (2) increasing blue carbon ranges from \$1.72 to \$3.30 per percent increase, and (3) increasing the number of jobs ranges from \$0.67 to \$0.98 per job. All else equal, WTP to avoid the status quo ranges from \$110.56 to \$238.46 per year.

In a recent study of Scotland's saltmarsh ecosystems, Riegel et al. (2023), find a non-monotonic annual marginal WTP response to increasing levels of blue carbon, first increasing from roughly £2.85 for a 1% increase in blue carbon (which is close to our values) to £-1.22 for a 2.5% increase, then decreasing to £-8 for a 5% increase. These

results from Riegel et al. (2023) are noticeably larger than ours for increases in MR size and blue carbon when looking to diminishing marginal utility in carbon storage.

Conversely, reductions in the marine system size are strongly disliked. We find that the magnitude of disutility from diminishment of the marine reserve system exceeds the utility of equivalent gains, a pattern consistent with the concept of loss aversion under prospect theory, and evidence of WTP asymmetries in environmental valuation (Kahneman Tversky 1979; Loomis et al., 1998). For instance, the effect on WTP of a 50% decrease in the size of the marine reserve system is noticeably larger in magnitude than the effect of a 50% increase in size. Even more pronounced is the relationship between decreases and increases in the number of jobs, where the magnitude of loss aversion is almost $150 per year.

We also find that survey participants on average are willing to pay annual amounts of between $465 and 704$ for an "optimistic scenario," whereby the size of the state's marine reserve system increases by 50%, 200 new jobs are created, and blue carbon increases by 100%. The upper end of this interval ($704), exceeds the study's highest program cost level of $500, indicating a potential face-validity problem (Glenk et al., 2024). To the contrary, participants are willing to pay between $237 and $375 to avoid a "pessimistic scenario," whereby the marine reserve system decreases by 50%, 200 jobs are lost, and there is no increase in blue carbon.

Respondents are willing to trade off a larger marine reserve system for a smaller level of blue carbon, as well as a larger number of jobs for a smaller level of blue carbon. However, the value of the marine reserve system increases with increases in the number of jobs, and the value of blue carbon increases with increases in the costs associated with the policy change. This latter result suggests a price-signaling effect, whereby a higher program cost signals higher value in blue carbon to the typical survey respondent.

Among the attributes examined, blue carbon has emerged as the most highly valued. Respondents have displayed reduced price-sensitivity when blue carbon benefits are high, although marginal utility declines at higher levels - suggesting diminishing returns. Together, these findings indicate that although people are willing to pay more for higher blue carbon benefits, the extra value they assign to each additional unit of blue carbon decreases. Still, when blue carbon gains are large, respondents are more willing to accept higher policy costs. Previous studies have found scope sensitivity in marine ecosystems (Jobstvogt et al, 2014), as well as insensitivity to scope (Remondou et al., 2015; Tyllianakis, 2022), which suggests that this pattern may depend upon the type of ecosystem services provided by marine ecosystems, features present within the reserves, and the public's perceptions of the benefits provided by ecosystem services. To our knowledge, Tyllianakis (2022) and Riegel et al. (2023) are the only previous studies that evaluate the value of additional blue carbon. Tyllianakis estimates a WTP of $118 for a 25% increase in blue carbon.

We find that respondents value job creation, including in coastal and marine contexts (Xuan, et al., 2021), a result consistent with the idea that linking environmental initiatives to local economic development can enhance public appeal. In practical terms, this finding aligns with the notion that environmental policies can win broader support if they are packaged with economic benefits, a point often emphasized in the literature on a just transition - fair and inclusive new jobs, training and support for workers and communities affected by environmental policies (ILO,2015).

In the face of constrained budgets, a blue carbon-focused package (large blue carbon gains with smaller increases in MR system size and coastal jobs) garners more support than an MR/job–focused package (larger increases in MR system size and coastal jobs, but smaller blue carbon gains). This pattern suggests that blue carbon benefits can offset lower economic co-benefits. Our survey results indicate that Oregonians prioritize the conservation and restoration of blue carbon habitat above the expansion of marine reserves. This WTP for blue carbon habitat expansion corroborates recent evidence from choice experiments showing positive WTP for blue carbon capture (Riegel et al., 2023; Tyllianakis, 2022) and habitat restoration as kelp forests (Hynes et al., 2021) and saltamarsh management (Grilli et al., 2022).

Kim et al (2022) find that the restoration of blue carbon ecosystems are more valued in tidal flats, seagrass meadows, and salt marshes (per unit areas restored). Cost-curve analyses rank tidal-marsh and seagrass restoration among the least-cost blue carbon options (McKinsey & Company, 2022), reinforcing the cost-effectiveness of prioritizing blue carbon targets. From a management perspective, these studies find that prioritizing blue-carbon targets creates cost-effective outcomes, and, given preference heterogeneity, differentiated-fee structures may be implemented to ensure fairness and effectiveness (Bergquist et al., 2022; Mohammadzadeh Valencia et al., 2024).

Recreational fishers, on average, exhibit lower mean WTP for MRS expansion than non-fishers yet they demand greater compensation to tolerate the status quo. Recreational fishers might favor improvements that directly enhance their experience, such as better enforcement against illegal fishing (Hall et al., 2002), but resist measures that unduly limit their activities such as encroachment of commercial fishing efforts. This nuanced stance is consistent with literature distinguishing extractive from non-extractive stakeholders (Johnston et al., 2010; Arlinghaus et al., 2019). Our finding that coastal recreationists assign a larger premium to blue carbon benefits than do non-recreationists aligns with evidence that personal experience with an ecosystem can affect its perceived value (Czajkowski et al., 2015).

Our stratification analysis further shows that respondents engaged solely in general ocean recreation (excluding fishing) place greater value on blue carbon benefits than those who also fish recreationally. Meanwhile, those who engage exclusively in recreational fishing express lower WTP for ocean-related employment gains than non-recreationists,

aligning with findings of existence value – that people are willing to pay for preservation even if they never visit the resource (Kniivilä, 2006).

Our findings also concur with recent findings presenting mixed evidence on nearshore spatial management valuation: while some studies report higher use values compared to non-use values for coastal and marine ecosystems (Tyllianakis et al., 2019), others find the opposite (Hynes et al., 2021). The divergence between these various groups has policy relevance: it suggests that optimal policy design might involve differentiated strategies or complementary programs. For instance, in Oregon, recreational and commercial fishers were heavily engaged in the marine reserve planning process, and had direct influence on the spatial boundaries to account for some of their specific preferences. Several marine reserve community partners have evolved into non-profit organizations that coordinate citizen science efforts near the reserves. These partners report that these engagement efforts have increased compliance, while simultaneously educating the broader public about tangible local benefits. Such tangible local benefits have the potential to sustain widespread support for these no-take areas.

From a methodological standpoint, our research underscores the value of employing advanced discrete choice models in capturing complex behavioral phenomena. The comparison of utility-space vs. WTP-space models, inclusion of error components, and modeling of scale heterogeneity are not just statistical exercises – they materially affect the interpretation of our data. For instance, the WTP-space RPL-EC model yields more reasonable and consistent welfare estimates than a corresponding utility-space model, which aligns with the recommendations of Train and Weeks (2005) and others that WTP-space estimation can reduce extreme value problems.

Traditional models assume that choices reflect clear preferences, but in reality respondents often express uncertainty, particularly when it comes to unfamiliar or new situations. We find that accounting for this respondent uncertainty can meaningfully change the results, revealing stronger underlying WTP for improvements, and moderating status quo aversion. This finding is intriguing in light of previous works. An early contingent valuation study by Champ et al. (1997) treated uncertainty by recoding "yes" responses to "no" if the respondent lacked confidence in their responses to a dichotomous-choice bid value, leading to more conservative WTP estimates. Lundhede et al. (2009) and Li and Ando (2023) adjusted choices for levels of uncertainty and found no significant effect on any of their attributes' marginal WTP estimates, except for the size of the status quo coefficient. In our case, accounting for respondent uncertainty did not simply dampen values; instead, it nuanced them (more boosting than reducing the coefficient estimates). Our results suggest that after controlling for response uncertainty, respondents place higher values on environmental improvements. This could reflect a higher level of knowledge or emotional investment in the issue among more certain respondents (e.g., perhaps climate-conscious individuals know better what they want from blue carbon pol-

icy). The implication for researchers is that gathering and utilizing such self-reported uncertainty measures can potentially improve welfare estimates' validity.

We find some evidence of attribute non-attendance (ANA) among our survey respondents. A model controlling for potential ANA in the status quo and cost attributes reports a larger WTP to avoid the status quo, but lower WTP estimates for the remaining attributes. However, statistical tests indicate that results obtained from our parsimonious linear model better explain the data generating process than those obtained from the model controlling for potential ANA. These results are corroborated by a comparison of standard goodness-of-fit measures. Hence, we interpret our ANA results with caution.

From a policy standpoint, our findings suggest both opportunities and challenges in changing Oregon's nearshore spatial management with blue carbon goals in mind. In particular, the strong preference for expanding blue carbon habitat indicates that framing coastal management as a climate action - not just management of fisheries or biodiversity - may broaden public support and increase policy appeal. This resonates with calls to integrate coastal ecosystems into climate strategies (e.g. inclusion of blue carbon in national and subnational climate plans, see Wedding et al., 2021) and to leverage these ecosystems in carbon credit or climate finance mechanisms. However, the distribution of benefits and costs suggests there will be winners and losers. Benefits from enhanced carbon sequestration, recreation, and climate resilience may be more salient to coastal residents and non-extractive users, while certain groups - especially fishers - may perceive marine or estuarine protected area expansions as restrictive. Policymakers must therefore anticipate stakeholder resistance and consider mitigation strategies, including participatory planning, and adaptive management. Likewise, funding mechanisms must be perceived as fair: flat taxes may face opposition from non-users, while opt-in conservation fees may achieve higher compliance among environmentally motivated individuals.[31]

While respondents value job creation as a result of marine and coastal policy changes, our results suggest that public support can remain high even when economic co-benefits are modest - provided ecological and carbon sequestration goals are met. However, explicitly communicating these co-benefits (including restoration jobs) can enhance support by appealing to both environmental and socioeconomic priorities. Im-plementing climate-beneficial nearshore spatial management policy will also require robust monitoring and accounting systems to ensure measurable and verifiable carbon gains if such benefits are to be monetized or included in official climate inventories.

While our results are varied and extend the existing literature on valuation of marine resources, they are clearly not exhaustive. The issue of blue carbon is currently understudied as a potential means to combat climate change on a large scale. Estimating

[31] In Oregon, state and federal government agencies may support pilot projects to expand blue carbon habitat using taxpayer funds, but the funds would likely come from the Oregon general fund, which expands and contracts dramatically depending on the economy. Also, funds may become available through private firms purchasing carbon offsets and land trusts acquiring more estuary habitat.

the number of new jobs that could be created to expand blue carbon habitats is likewise a more complicated undertaking than assumed in this study. There are also a host of methodological issues that this study leaves unaddressed. For example, our welfare estimates are potentially biased upward due to the non-existence of what Kilders and Caputo (2023) call a “reference price” for the attributes evaluated in this study. The authors show that including respondent-specific reference prices in the interval of cost levels of a private good (in their case the ribeye meat market) can lead to more conservative WTP estimates than in traditional discrete choice designs that do not include these prices. Naturally, the challenge for us is how to incorporate relevant reference prices in a choice experiment designed to elicit preferences for non-marketed environmental goods.

A key limitation with our study is that the DCE described blue carbon benefits as percentage changes (0–100%) without pairing them with simple, relatable comparisons that people can immediately understand (for example, "equivalent to taking X cars off the road"). Our approach was consistent with other studies (Bocci et al. 2020; Brodersen et al. 2024) and a result of the lack of a blue carbon baseline in Oregon territorial waters, which stems from the complexity and uncertainty in calculating carbon sequestration (Börger et al. 2014; Luisetti et al. 2014; Macreadie et al. 2019). Nonetheless, respondents may understand that "more is better" yet struggle to judge the size of a 50% versus 100% increase. This discrepancy likely added noise to choices involving the blue carbon attribute. The present results provide an initial assessment of public preferences for blue carbon in Oregon. In future research, it is suggested to pair the percentage change of blue carbon with comparisons, such as car emission equivalents. A one-question comprehension check in the pilot test can be used to assess respondent grasp of the meaning of the blue carbon comparisons before proceeding.

Future work could also integrate revealed preference data to validate or complement our stated preference results. For instance, researchers might conduct a hedonic study to examine if coastal property markets capitalize proximity to coastal environments. The estimates from these types of studies could then be compared with the WTP estimates obtained from our study. Finally, we recommend that future choice experiments on environmental goods routinely consider more detailed subsampling by relevant use or demographic variables, or employ latent class and hybrid choice models to uncover distinct preference patterns (Boxall & Adamowicz, 2002).

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Tables and Figures

Table 1: Choice experiment attributes (and corresponding variable names)

- **Marine Reserve System (*MR*)**
  - No change in the size of the marine reserve system.
  - 50% decrease in the size of the marine reserve system.
  - 50% increase in the size of the marine reserve system.
- **Blue Carbon (*Carbon*)**
  - No change in carbon capacity sequestration.
  - 25% increase in carbon sequestration.
  - 50% increase in carbon sequestration.
  - 100% increase in carbon sequestration.
- **Coastal Jobs (*Jobs*)**
  - No change in number of jobs.
  - 200 jobs lost.
  - 200 jobs gained.
- **Annual Household Cost (*Cost*)**
  - $0
  - $25
  - $50
  - $100
  - $200
  - $400
  - $500

Note: Variable names are provided in parentheses (in italics). The "no change" and "$0" levels for each attribute appeared in some choice alternatives and were the levels assigned to the status quo.

Table 2: Socio-demographic variables, sample percentages for the "=1" category, and U.S. census estimates for Oregon.

Variable	**Description**	**% of Sample**[a]	**Census Estimate**[b]
gender[c]	=1 if male, 0 otherwise	61.2% (56.7, 65.7)	49.7%
senior	=1 if age is 55+ years, 0 otherwise	38.3% (33.8, 42.8)	31.5%
midage	=1 if age 35–54, 0 otherwise	40.4% (35.8, 45.0)	40.6%
white	=1 if identifies as White, 0 otherwise	89.3% (86.4, 92.2)	73.9%
latino	=1 if Hispanic/Latino, 0 otherwise	7.48% (5.02, 9.93)	14.9%
black	=1 if Black/African-American, 0 otherwise	2.49% (1.03, 3.94)	2.10%
native	=1 if Amer. Indian/Alaska Native, 0 otherwise	2.72% (1.20, 4.23)	1.10%
asian	=1 if Asian, 0 otherwise	4.31% (2.41, 6.20)	4.6%
pacific	=1 if Nat. Hawaiian/Pac. Isl., 0 otherwise	0.68%(–0.09, 1.45)	0.5%
other	=1 if Other race, 0 otherwise	3.63% (1.88, 5.37)	2.9%
college[d]	=1 if some college/Associate, 0 otherwise	36.51%(32.0, 41.0)	32.3%
bachdeg[d]	=1 if Bachelor's degree, 0 otherwise	29.3% (25.0, 33.5)	23.5%
graddeg[d]	=1 if Graduate degree, 0 otherwise	16.3% (12.9, 19.7)	14.2%
highinc[e]	=1 if income $250k, 0 otherwise	3.40% (1.71, 5.09)	11.8%
himidinc	=1 if $100k income <$250k, 0 otherwise	28.1% (23.9, 32.3)	27.8%
midinc	=1 if $50k income <$100k, 0 otherwise	42.6% (38.0, 47.2)	29.5%
workft[f]	=1 if working full-time, 0 otherwise	47.2% (42.5, 51.8)	60.0%
workpt	=1 if working part-time, 0 otherwise	10.2% (7.37, 13.0)	—
homemaker	=1 if homemaker, 0 otherwise	2.27% (0.88, 3.66)	37.4%
student[g]	=1 if student, 0 otherwise	1.81% (0.56, 3.05)	—
retired	=1 if retired, 0 otherwise	19.0% (15.4, 22.7)	—
urban[h]	=1 if urban/peri-urban area, 0 otherwise	27.9% (23.7, 32.1)	67.0%

Notes:

a 95% confidence intervals in parenthesis.
b Estimates obtained from the U.S. Census Bureau, American Community Survey, 2023, one year data profiles.
https://data.census.gov/table/ACSDP1Y2023.DP05?g=040XX00US41 and
https://data.census.gov/profile/Oregon?g=040XX00US41#education.
c Respondents were able to identify as "other" or "prefer not to say". However, none chose either of these two personal identifiers.
d Population 25 years and older.
e Census estimate for `highinc` is $200,000 or greater annual household income, and for `himidinc` is at least $100,000 and no greater than $200,000.
f Population over 16 years of age employed in civilian labor force and armed services.
g Population over 16 years of age not in labor force.
h Obtained from https://www.ohsu.edu/oregon-office-of-rural-health/about-rural-and-frontier-data.

Table 3: Linear Model Results

	RPL		**RPL-EC**		**RPL-EC -2 (WTP-Space)**†		**GMXL (WTP-Space)**†	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
SQ	-0.994^{***}	2.134^{***}	-1.003^{***}	2.101^{***}	-110.556^{***}	333.521^{***}	-238.463^{***}	723.382^{***}
	(0.190)	(0.265)	(0.202)	(0.595)	(34.964)	(84.995)	(66.756)	(86.984)
MR	0.019^{***}	0.019^{***}	0.019^{***}	0.019^{***}	2.051^{***}	4.002^{***}	3.568^{***}	3.578^{***}
	(0.003)	(0.004)	(0.003)	(0.004)	(0.393)	(0.674)	(0.456)	(0.720)
Jobs	0.004^{***}	0.003^{***}	0.004^{***}	0.003^{***}	0.671^{***}	0.369^{***}	0.982^{***}	0.624^{***}
	(0.001)	(0.001)	(0.001)	(0.001)	(0.075)	(0.141)	(0.116)	(0.187)
Carbon	0.010^{***}	0.001	0.010^{***}	0.001	2.747^{***}	4.179^{***}	3.296^{***}	4.152^{***}
	(0.002)	(0.010)	(0.002)	(0.007)	(0.464)	(0.637)	(0.548)	(0.820)
Cost	-0.006^{***}	0.013^{**}	-0.006^{***}	0.012	-0.0046^{***}	0.004^{***}	-1	–
	(0.001)	(0.007)	(0.001)	(0.015)	(0.0004)	(0.0005)		
Scale heterogeneity τ	–	–	–	–	–	–	0.268	–
							(0.497)	
Scale sample mean $-\sigma$	–	–	–	–	–	–	0.997^{***}	–
							(0.268)	
Error component μ	–	–	–	0.462	–	1.988^{***}	–	–
				(2.626)		(0.199)		
Log-likelihood	$-1{,}299.95$		$-1{,}299.93$		$-1{,}325.16$		$-1{,}339.51$	
Pseudo-R^2	0.206		0.206		0.187		0.182	
AIC/N	1.779		1.780		1.801		1.819	
BIC/N	1.850		1.855		1.840		1.854	
HIC/N	1.806		1.808		1.815		1.832	
# parameters	20		21		11		10	

Note: † Model estimated with uncorrelated parameters. Standard errors in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% levels.

Table 4: Marginal WTP Estimates for Attributes from Linear Models

	RPL	**RPL-EC**	**RPL-EC -2 (WTP-Space)**†	**GMXL (WTP-Space)**†
SQ	-165.789^{***}	-168.329^{***}	-110.556^{***}	-238.463^{***}
	(36.430)	(37.332)	(34.964)	(66.756)
MR	3.196^{***}	3.213^{***}	2.051^{***}	3.568^{***}
	(0.396)	(0.396)	(0.393)	(0.456)
Jobs	0.665^{***}	0.670^{***}	0.671^{***}	0.982^{***}
	(0.098)	(0.103)	(0.075)	(0.117)
Carbon	1.718^{***}	1.735^{***}	2.747^{***}	3.296^{***}
	(0.340)	(0.367)	(0.464)	(0.548)

Note: †Model estimated with uncorrelated parameters. Standard errors are in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% levels, respectively.

Table 5: Non-Linear Models Results

	RPL		**RPL-EC**		**RPL-EC 2 (WTP-Space)†**	
	Mean	SD	Mean	SD	Mean	SD
SQ	-0.044	0.477	-1.946	11.232***	-155.030**	12.731
	(0.593)	(0.813)	(3.882)	(3.607)	(68.786)	(157.689)
MR (-50%)	-2.908***	2.568*	-23.449***	12.175***	-204.357***	225.398**
	(0.945)	(1.336)	(6.879)	(4.197)	(47.585)	(101.895)
MR (+50%)	1.607**	1.651	8.276**	27.317***	87.112*	204.593*
	(0.676)	(1.758)	(4.214)	(7.215)	(44.765)	(108.960)
Jobs (-200)	-1.504**	2.714***	-15.735***	9.311**	-139.820***	121.234
	(0.670)	(1.030)	(5.184)	(4.647)	(45.395)	(103.511)
Jobs (+200)	1.341**	0.116	4.861	12.731**	138.141**	133.628
	(0.584)	(0.946)	(3.703)	(5.749)	(59.466)	(112.672)
Carbon (+25%)	3.690***	0.586	25.663***	6.383*	277.724***	504.436***
	(1.142)	(1.319)	(7.060)	(3.802)	(74.587)	(113.371)
Carbon (+50%)	3.543***	2.099*	25.213***	22.595***	275.671***	323.894***
	(1.130)	(1.249)	(7.409)	(7.272)	(67.750)	(100.017)
Carbon (+100%)	3.427***	1.514	25.596***	0.475	383.165***	702.967***
	(1.094)	(1.910)	(7.865)	(4.982)	(70.782)	(125.407)
Cost	-0.0130***	0.006	-0.106***	0.141***	-0.004***	0.002***
	(0.004)	(0.014)	(0.029)	(0.044)	(0.0005)	(0.0005)
Error component - μ				22.819***		1.492***
				(6.899)		(0.189)
Log-likelihood	-1269.384		-1259.345		-1303.495	
Pseudo-R^2	0.225		0.231		0.200	
AIC/N	1.784		1.771		1.782	
BIC/N	1.976		1.968		1.850	
HIC/N	1.855		1.845		1.808	
#parameters	54		55		19	

Note: †Model estimated with uncorrelated parameters.
Standard errors are in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% levels, respectively.
The uncorrelated GMXL model fits the data poorly in this specification, showing no significant coefficient estimates for any of the attributes, as well as an overwhelmingly highly significant tau scale. Hence, we decided to drop this model from the analyses of this specification.

Table 6: Marginal WTP Estimates for Attributes from Non-Linear Models

	RPL	RPL-EC	RPL-EC 2 (WTP-Space)†
SQ	3.372	-18.425	-155.030**
	(45.689)	(36.201)	(68.786)
MR (-50%)	-223.996***	-221.998***	-204.357***
	(43.751)	(33.416)	(47.585)
MR (+50%)	123.807***	78.351**	87.119*
	(43.576)	(35.962)	(44.765)
Jobs (-200)	-115.845***	-148.967***	-139.820***
	(40.477)	(34.567)	(45.395)
Jobs (+200)	103.338**	46.018	138.141**
	(48.536)	(32.720)	(59.466)
Carbon (+25%)	284.276***	242.962***	277.724***
	(57.399)	(39.482)	(74.587)
Carbon (+50%)	272.925***	238.702***	275.671***
	(53.277)	(38.639)	(67.750)
Carbon (+100%)	263.986***	242.327***	383.165***
	(54.679)	(42.120)	(70.782)

Note: †Model estimated with uncorrelated parameters.
Standard errors are in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% level, respectively.

Table 7: Linear Models Results with Attribute Interaction Terms

	RPL†		RPL-EC†	
	Mean	SD	Mean	SD
SQ	-0.922***	2.403***	-0.92491***	2.298***
	(0.199)	(0.223)	(0.198)	(0.563)
MR	0.0204***	0.011***	0.020***	0.011***
	(0.003)	(0.002)	(0.003)	(0.002)
Jobs	0.005***	0.002***	0.005***	0.002***
	(0.0007)	(0.0007)	(0.0007)	(0.0007)
Carbon	0.007***	0.011***	0.007***	0.011***
	(0.002)	(0.003)	(0.002)	(0.003)
Cost	-0.004***	0.004***	-0.004***	0.004***
	(0.0007)	(0.0007)	(0.0007)	(0.0007)
MR × Jobs	0.00003**		0.00003**	
	(0.00001)		(0.00001)	
MR × Carbon	-0.0001***		-0.0001***	
	(0.00005)		(0.00005)	
MR × Cost	-0.00001		-0.00001	
	(0.00001)		(0.00001)	
Jobs × Carbon	-0.000038***		-0.00004***	
	(0.000014)		(0.00001)	
Jobs × Cost	-0.0001		-0.0000009	
	(0.000003)		(0.000003)	
Carbon × Cost	0.00002**		0.00002**	
	(0.00001)		(0.00001)	
Error Component μ				0.660
				(1.764)
Log-likelihood	-1322.261		-1322.608	
Pseudo-R^2	0.192		0.192	
AIC/N	1.802		1.804	
BIC/N	1.856		1.861	
HIC/N	1.822		1.825	
# parameters	15		16	

Note: †Model estimated with uncorrelated parameters.
Standard errors are in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% level, respectively.

Table 8: WTP Estimates for Optimistic, Status Quo and Pessimistic Scenarios

	RPL	RPL-EC	RPL-EC -2 (WTP-Space)†	GMXL (WTP-Space)†
Optimistic	464.646***	468.210***	511.515***	704.428***
	(54.561)	(58.245)	(66.814)	(74.354)
SQ	-165.789***	-168.329***	-110.556***	-238.463***
	(36.430)	(37.332)	(34.964)	(66.756)
Pessimistic	-292.898***	-294.748***	-236.840***	-374.857***
	(34.626)	(34.770)	(29.305)	(36.113)

Note: †Model estimated with uncorrelated parameters.
WTP scenarios are estimated by Delta Method with 2,000 draws.
Standard errors are in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% level, respectively.

Table 9: Linear Model Results by Subsample Related to Fish Recreation on the Oregon Coast

	Fishing Recreation (n=189)				**No-Fishing Recreation (n=159)**			
	RPL-EC (WTP-Space)		RPL-EC -2 (WTP-Space)†		RPL-EC 1 (WTP-Space)		RPL-EC -2 (WTP-Space)†	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
SQ	-89.554^{a,b}	30.402	-167.308^{***a,b}	9.918	-23.723	24.055	-106.145**	0.914
	(62.925)	(59.850)	(58.596)	(94.144)	(98.715)	(110.137)	(48.022)	(52.972)
MR	-3.153^{b}	19.886***	2.984^{***b}	2.291***	4.637***	0.973*	4.503***	2.111***
	(1.933)	(7.554)	(0.565)	(0.620)	(1.010)	(0.563)	(0.564)	(0.599)
Jobs	-0.482^{b}	0.739^{**b}	0.714^{***b}	0.617***	0.693***	0.175**	0.980***	0.495***
	(0.485)	(0.340)	(0.147)	(0.146)	(0.228)	(0.088)	(0.127)	(0.154)
Carbon	2.308^{b}	7.675^{b}	2.704^{***b}	2.549***	4.134***	1.792**	4.613***	4.120***
	(2.488)	(7.331)	(0.657)	(0.490)	(1.183)	(0.845)	(0.696)	(0.810)
Cost	-0.007***		-0.003***	0.003***	-0.010**	0.002	-0.005***	0.003***
	(0.002)		(0.0005)	(0.0007)	(0.005)	(0.002)	(0.0007)	(0.0008)
Error component - μ		2.314***		2.183***		2.392***		2.270***
		(0.514)		(0.470)		(0.694)		(0.480)
Log-likelihood	-675.182		-691.01		-515.956		-531.974	
Pseudo-R2	0.187		0.168		0.261		0.239	
AIC/N	1.842		1.857		1.689		1.707	
BIC/N	1.970		1.925		1.836		1.785	
HIC/N	1.891		1.883		1.746		1.737	
#parameters	21		11		21		11	

Note: †Model estimated with uncorrelated parameters. Standard errors are in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% level, respectively.
"a" denotes significance of the differences between "Fishing Recreation" and "No-Fishing Recreation" WTP estimates across respective correlated and uncorrelated specifications at the 5% level for the convolution test.
"b" denotes significance of the differences between subsample's WTP estimates across respective correlated and uncorrelated specifications at the 1% level for the Mann-Whitney-U test.

Table 10: Linear Model Results by Subsamples Related to General Recreation in the Oregon Coast

	General Recreation (n=230)				**No-General Recreation (n=118)**			
	RPL-EC 1 (WTP-Space)		RPL-EC -2 (WTP-Space)†		RPL-EC 1 (WTP-Space)		RPL-EC -2 (WTP-Space)†	
Attribute	Mean	SD	Mean	SD	Mean	SD	Mean	SD
SQ	-62.967^{b}	7.763	-139.982^{***b}	64.575	-80.011	41.721	-185.538**	166.433***
	(39.646)	(22.440)	(44.317)	(44.495)	(71.062)	(80.409)	(81.906)	(62.158)
MR	1.789^{***b}	2.209***	3.230^{***b}	2.340***	2.090***	1.233**	3.657***	3.206***
	(0.393)	(0.347)	(0.486)	(0.421)	(0.625)	(0.506)	(0.818)	(0.881)
Jobs	0.384^{***b}	0.142	0.726^{***b}	0.386***	-0.018	0.393	1.032***	0.361**
	(0.148)	(0.312)	(0.105)	(0.092)	(0.490)	(0.843)	(0.179)	(0.172)
Carbon	4.138^{***b}	3.620**	1.750^{***b}	1.707***	-2.451	1.590	1.330	5.509***
	(1.196)	(1.843)	(0.602)	(0.483)	(3.840)	(4.082)	(1.194)	(1.438)
Cost or ϑ	-0.006***	0.006***	-0.004***	0.005***	-0.002	0.012	-0.004***	0.003***
	(0.002)	(0.002)	(0.001)	(0.0007)	(0.005)	(0.009)	(0.0006)	(0.0007)
Error component - μ		2.012***		2.216***		1.785***		2.811***
		(0.404)		(0.290)		(0.376)		(0.593)
Log-likelihood	-802.885		-823.119		-422.187		-406.46	
Pseudo-R2	0.206		0.186		0.186		0.216	
AIC/N	1.791		1.813		1.878		1.769	
BIC/N	1.901		1.871		2.063		1.866	
HIC/N	1.833		1.835		1.951		1.807	
#parameters	21		11		21		11	

Note: †Model estimated with uncorrelated parameters.
Standard errors are in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% level, respectively.
"a" denotes significance of the differences between "General Recreation" and "Non-General Recreation" WTP estimates across respective correlated and uncorrelated specifications at the 5% level for the convolution test.
"b" denotes significance of the differences between subsample's WTP estimates across respective correlated and uncorrelated specifications at the 1% level (Mann-Whitney-U test).

Table 11: Ex post stated ANA results – WTP estimates.

Attribute	**RPL**	**RPL-EC**
SQ	-123.02^{***}	-125.25^{***}
	(33.17)	(33.85)
MR	2.79^{***}	2.77^{***}
	(0.41)	(0.43)
Carbon	1.01^{***}	1.01^{***}
	(0.36)	(0.38)
Jobs	0.70^{***}	0.70^{***}
	(0.10)	(0.11)
$MR \times ANA$	-1.69^{*}	-1.64^{**}
	(0.94)	(0.91)
$Carbon \times ANA$	4.69^{***}	4.68^{***}
	(1.21)	(1.20)
$Jobs \times ANA$	-0.31	-0.31
	(0.20)	(0.20)
μ	—	1.98^{**}
		(0.95)
log-likelihood	$-1{,}296.06$	$-1{,}295.26$
χ^2	682.10^{***}	683.69^{***}
pseudo-R^2	0.21	0.21
AIC	1.766	1.766
BIC	1.816	1.819
HIC	1.784	1.786

Notes: Standard errors in parentheses, calculated via the Delta Method with 2,000 random draws. *** significant at 1%, ** at 5%, * at 10%. Both models estimated with uncorrelated parameters. WTP for interaction terms is computed with *cost*ana* included in the denominator of the WTP expression.

Table 12: Ex post implied ANA results – WTP estimates.

Attribute	**WTP**
SQ	411.47***
	(89.91)
MR	1.55***
	(0.22)
Carbon	1.05***
	(0.29)
Jobs	0.40***
	(0.05)
log-likelihood	−1,637.11
χ^2	470.96***
pseudo-R^2	0.14
AIC	1.898
BIC	1.923
HIC	1.908

Notes: Standard errors in parentheses, calculated using the Delta Method with 2,000 random draws. *** significant at 1%, ** at 5%, * at 10%. Model estimated with fixed parameters.

Figure 1: Map of Oregon's Marine Reserve System

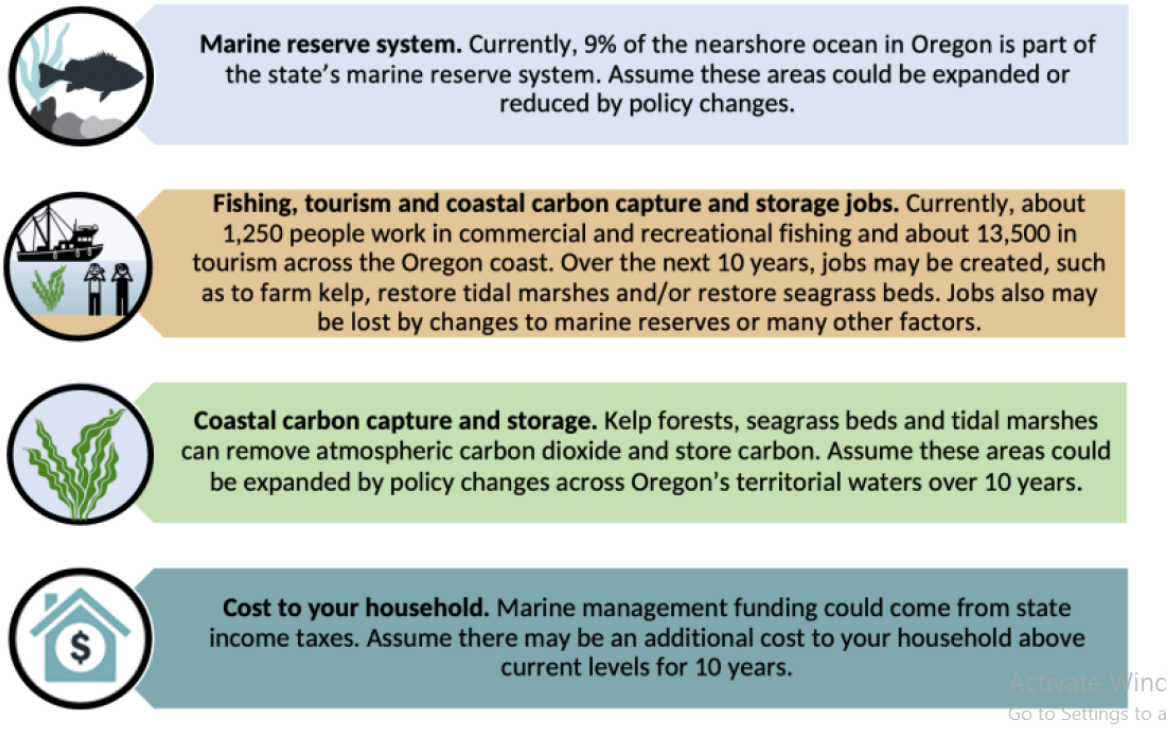
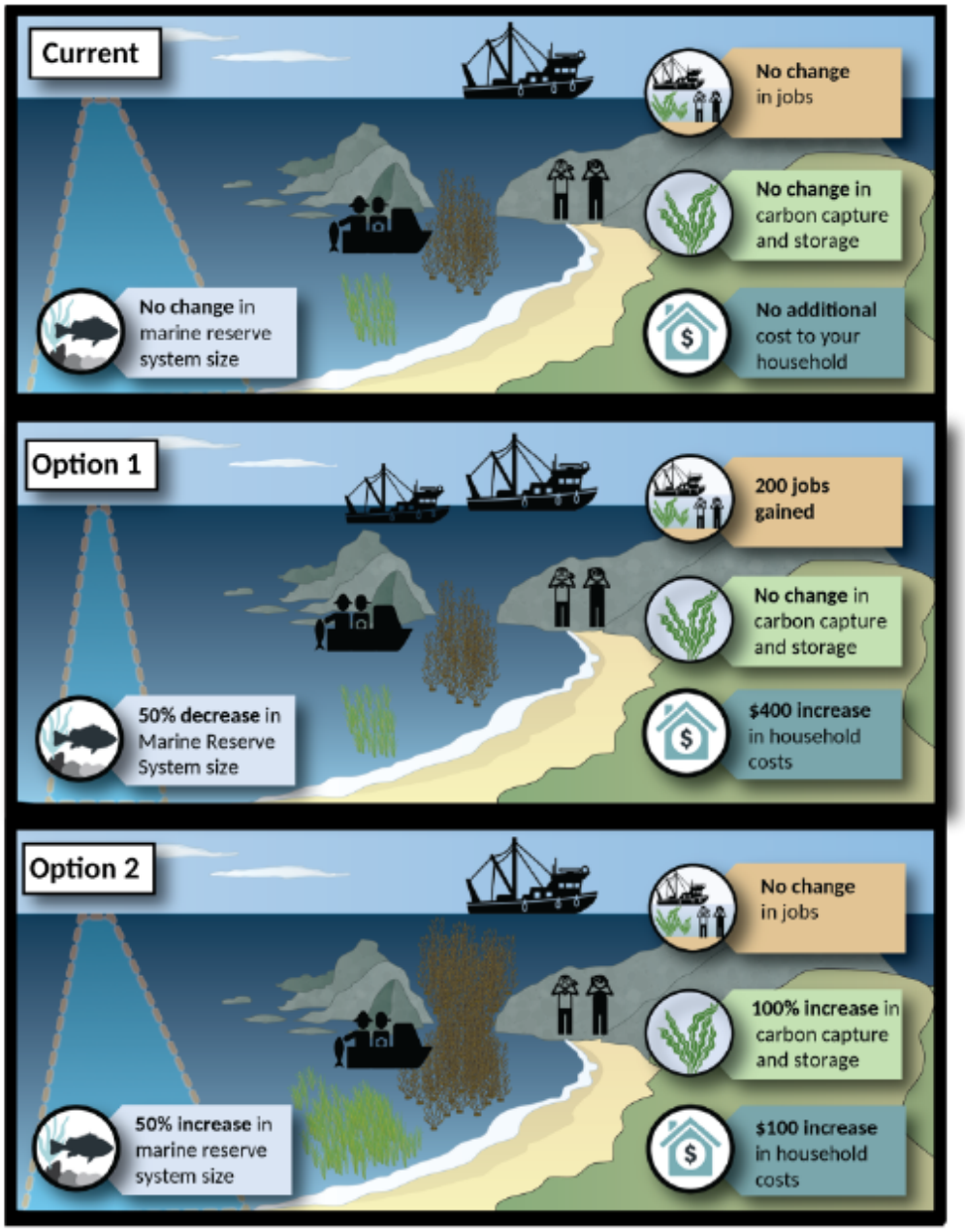


Figure 2: Panel A: Attribute description shown to respondents.



- ◯ Current situation (no change)
- ◯ Option 1
- ◯ Option 2

How certain are you that the choice you made in this comparison reflects your preferences?

0 = Not at all certain; 100 = Completely certain

0 10 20 30 40 50 60 70 80 90 100

Figure 2: Panel B: Description of the choice tasks shown to respondents.

Appendices

Appendix A

Table A. Parsimonious Linear Model Results by Subsample (Utility space)

	Fish Recreation (n=189)				**No-Fish Recreation** (n=159)			
	RPL†		RPL-EC†		RPL†		RPL-EC†	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
SQ	-0.702***	2.179***	-0.709***	1.326	-0.792***	2.306***	-0.798**	2.243
	(0.236)	(0.287)	(0.257)	(6.479)	(0.285)	(0.341)	(0.336)	(1.991)
MR	0.009***	0.010***	0.009***	0.010***	0.013***	0.012***	0.013***	0.012***
	(0.002)	(0.003)	(0.002)	(0.004)	(0.002)	(0.004)	(0.003)	(0.003)
Jobs	0.003***	0.002**	0.003***	0.002**	0.004***	0.001	0.004***	0.001
	(0.0005)	(0.001)	(0.0005)	(0.0009)	(0.0006)	(0.001)	(0.0006)	(0.001)
Carbon	0.006***	0.013***	0.006***	0.013***	0.017***	0.012***	0.017***	0.012***
	(0.002)	(0.004)	(0.002)	(0.004)	(0.003)	(0.005)	(0.003)	(0.004)
Cost or ϑ	-0.003***	0.003***	-0.003***	0.003***	-0.004***	0.004***	-0.004***	0.004***
	(0.0005)	(0.0005)	(0.0004)	(0.0004)	(0.0007)	(0.0007)	(0.0006)	(0.0006)
Error component - mu				1.720		0.504		
				(5.083)				(7.896)
Log-likelihood	-688.607		-688.156		-560.036		-559.975	
Pseudo-R^2	0.152		0.153		0.230		0.230	
AIC/N	1.846		1.847		1.789		1.792	
BIC/N	1.901		1.908		1.852		1.862	
HIC/N	1.867		1.871		1.814		1.820	
# parameters	9		10		9		10	

†Model estimated with uncorrelated parameters. Standard errors in parentheses.
*, **, *** denote significance at the 10%, 5%, and 1% levels, respectively.

Comparing the coefficient estimates in utility space (Table A), we can't notice an apparently big difference between the subsample of people who engage and do not engage in fishing recreational activities. The error component was not significant for both models analyzed.

The WTP estimates in magnitude also are not quite different (Table B). However, the Mann-Whitney-U tests capture shifts in median between fishermen and no-fishermen subsamples, this can be explained mostly because of the small standard deviation obtained for all parameters in the utility-space specifications. In this case, because we have not estimated the WTP coefficients directly, the WTP estimates were calculated based on 10,000 independent bootstrap subsamples from the individual posterior WTP obtained directly from the model estimation in NLOGIT (Greene, 2016).We estimated the standard error of each mean WTP by drawing 10,000 bootstrap samples (with replacement) from the 10,000 simulated WTP draws, computing the mean for each sample, and then taking the standard deviation of those 10,000 bootstrap-means as the bootstrap standard error.

Table B. WTP Estimates by Subsample Related to Fish Recreation on the Oregon Coast (Utility space)

	Fish Recreation		**No-Fish Recreation**	
	RPL†	RPL-EC†	RPL†	RPL-EC†
SQ	-378.118***b	-368.007***b	-278.374***	-279.540***
	(61.917)	(24.764)	(45.683)	(43.611)
MR	4.474***b	4.472***b	4.297***	4.278***
	(0.194)	(0.200)	(0.209)	(0.208)
Jobs	1.365***b	1.368***b	1.232***	1.230***
	(0.034)	(0.037)	(0.030)	(0.030)
Carbon	3.233***b	3.202***b	5.761***	5.760***
	(0.262)	(0.269)	(0.240)	(0.239)

†Model estimated with uncorrelated parameters.
Standard errors are in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% levels, respectively.
[a]significance of differences between "Fishing Recreation" and "No-Fish Recreation" WTP estimates (convolution test, $\Pr(X > Y) > 0.5$) at 5% level.
[b]significance of differences between correlated vs. uncorrelated model estimates (Mann–Whitney U test) at 1% level.

Table C. Parsimonious Linear Model Results by Subsamples Related to General Recreation in the Oregon Coast (Utility Space)

	General Recreation (n=230)				No-General Recreation (n=118)			
	RPL†		RPL-EC†		RPL†		RPL-EC†	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
SQ	-0.660***	2.065***	-0.675***	0.966	-0.967***	2.650***	-0.981**	2.254
	(0.210)	(0.255)	(0.242)	(5.543)	(0.359)	(0.437)	(0.419)	(4.440)
MR	0.012***	0.010***	0.012***	0.010***	0.008***	0.012***	0.008***	0.0119***
	(0.002)	(0.003)	(0.002)	(0.003)	(0.002)	(0.004)	(0.003)	(0.004)
Jobs	0.003***	0.002***	0.003***	0.002***	0.003***	0.001	0.003***	0.001
	(0.0004)	(0.0008)	(0.0004)	(0.0007)	(0.0006)	(0.002)	(0.0006)	(0.002)
Carbon	0.010***	0.013***	0.010***	0.013***	0.012***	0.014***	0.012***	0.014***
	(0.002)	(0.003)	(0.002)	(0.003)	(0.003)	(0.005)	(0.003)	(0.005)
Cost or ϑ	-0.003***	0.003***	-0.003***	0.003***	-0.004***	0.004***	-0.004***	0.004***
	(0.0005)	(0.0005)	(0.0004)	(0.0004)	(0.0007)	(0.0007)	(0.001)	(0.001)
Picture			1.837	(2.857)			1.379	(7.836)
Log-likelihood	-835.21		-835.209		-417.591		-417.564	
Pseudo-R^2	0.174		0.174		0.21		0.21	
AIC/N	1.835		1.837		1.808		1.812	
BIC/N	1.882		1.890		1.887		1.900	
HIC/N	1.853		1.857		1.839		1.846	
#parameters	9		10		9		10	

Note: †Model estimated with uncorrelated parameters.
Standard errors in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% level, respectively.

Table C shows no apparent variation between utility space coefficients of general recreational subsamples. The RPL and RPL-EC versions are very similar, and the RPL-EC component was not significant in either of the group's specifications.

Table D. WTP Estimates by Subsamples Related to General Recreation in the Oregon Coast (Utility Space)

	Rec_gen		Rec_nogen	
	RPL†	RPL-EC†	RPL†	RPL-EC†
SQ	-319.579***b	-296.064***b	-362.590***	-366.914**
	(50.084)	(13.541)	(60.748)	(44.980)
MR	5.401***b	5.387***b	3.240***	3.250***
	(0.120)	(0.206)	(0.206)	(0.208)
Jobs	1.379***b	1.378***b	1.211***	1.212***
	(0.037)	(0.038)	(0.025)	(0.026)
Carbon	4.560***b	4.561***b	4.691***	4.687
	(0.257)	(0.259)	(0.267)	(0.270)

Note: †Model estimated with uncorrelated parameters. Standard errors are in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% level, respectively.
a denotes significance of the differences between General Recreation and No-General Recreation WTP estimates across respective correlated and uncorrelated specifications at the 5% level (convolution test).
b denotes significance of the differences between subsample WTP estimates across correlated and uncorrelated specifications at the 1% level (Mann-Whitney-U test). It checks for shifts in medians (or, more generally, the location) across the entire distribution, using ranks of the value, therefore testing if one group tends to have higher value than another $P(X > Y) > 0.5$

Analyzing the WTP differences in Table D, the group which engages in ocean's recreational activities presents a higher SQ, MR and Jobs coefficients confirmed by the Mann-Whitney U test. Carbon is slightly lower for this group, also confirmed from this test. No full significant stochastic difference in WTP distribution was detected by the convolution test.

Appendix B

Table E shows MNL results for subsamples of people who do not participate in fishing recreational activities plus engage or not engage in general ocean recreational activities. We can see that not being a recreational fisherman but doing other ocean's recreational activities fits better the data (lower AIC/N) than not doing any recreational activity.

Table E. MNL Parsimonious Linear Model Results by Subsamples Related to Exclusive General Recreation and No Recreational Involvement in the Oregon Coast

	No-Fish Recreation but General Recreation (n = 82)	No-Fish Recreation and No-General Recreation (n = 77)
SQ	-0.599	0.007***
	(0.166)	(0.002)
MR	0.011***	0.003***
	(0.002)	(0.0005)
Jobs	0.002***	0.011***
	(0.0004)	(0.002)
Carbon	0.015***	-0.00285***
	(0.002)	(0.0005)
Cost or ϑ	-0.003***	-0.168
	(0.0005)	(0.169)
Log-likelihood	-319.21	-307.461
AIC/N	1.977	2.029
#parameters	5	5

Note: Standard errors are in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% level, respectively.

The WTP estimates by Delta Method with 2,000 draws (Table F) show that people who solely engage in ocean recreational activities have a higher premium for increases in nearshore spatial management size and blue carbon capture compared with individuals who do not participate in any recreational activity. The significant differences in WTP imply that individuals who actively interact with coastal ecosystems are more likely to perceive direct or indirect benefits from marine protection and climate mitigation efforts. This reinforces the idea that personal experience and emotional connection to the marine environment strengthen support for conservation policies. Consequently, policy strategies that emphasize recreational access and ecosystem co-benefits may enhance public valuation and legitimacy of nearshore spatial management expansions and blue carbon initiatives.

Table F. MNL WTP Estimates by Subsamples Related to Exclusive General Recreation and No Recreational Involvement in the Oregon Coast

	No-Fish Recreation but General Recreation	No-Fish Recreation and No-General Recreation
SQ	-21.382	-59.094
	(58.821)	(58.237)
MR	3.969***	2.508***
	(0.793)	(0.718)
Jobs	0.852***	0.985***
	(0.212)	(0.221)
Carbon	5.323***	3.809***
	(1.030)	(0.898)

Note: Standard errors are in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% level, respectively.

Appendix C - Analyzing Subsamples with and without Choice Uncertainty

In the survey, participants were asked after each choice task to state how certain the choice they made reflects their preferences on a scale from zero to 100 percent. If the respondent selected any value lower than 50% in the scale, the original choice was modified to the "Status Quo" alternative (always the third alternative in the choice task). If the original choice was already the "Status Quo" the participant's response remained unchanged. In this way, we ran a model that could capture the level of uncertainty of the respondents regarding their choice of alternatives. We compare these results with the analogous model that does not account for respondent uncertainty, and check if controlling for uncertainty generates more or less conservative WTP estimates.

Table G. Linear Models Results Controlling for Choice Uncertainty

	RPL		**RPL-EC**		**RPL-EC 2 (WTP-Space)**†		**GMXL (WTP-Space)**†	
Attribute	Mean	SD	Mean	SD	Mean	SD	Mean	SD
SQ	-0.1450	2.162***	-0.119	2.000***	15.820	63.938***	105.915***	0.682
	(0.170)	(0.251)	(0.189)	(0.736)	(27.956)	(22.744)	(25.365)	(158.506)
MR	0.018***	0.017***	0.018***	0.017***	3.363***	0.279	3.683***	4.628***
	(0.003)	(0.004)	(0.002)	(0.004)	(0.418)	(0.629)	(0.487)	(0.611)
Jobs	0.004***	0.003***	0.004***	0.003***	0.718***	0.031	0.890***	0.094
	(0.0005)	(0.0008)	(0.0005)	(0.0008)	(0.098)	(0.166)	(0.103)	(0.302)
Carbon	0.009***	0.00004	0.009***	0.002	3.397***	3.952***	2.526***	8.023***
	(0.002)	(0.007)	(0.002)	(0.006)	(0.489)	(0.683)	(0.763)	(0.969)
Cost or ϑ	-0.006***	0.013***	-0.006***	0.009	-0.005***	0.004***	-1	
	(0.0009)	(0.004)	(0.0008)	(0.017)	(0.0005)	(0.0005)		
Scale heterogeneity - τ							1.494***	
							(0.276)	
Scale sample mean - σ							0.925	
							(1.859)	
Error component - μ				0.800		1.414***		
				(1.784)		(0.198)		
Log-likelihood	-1301.814		-1301.640		-1315.008		-1376.356	
Pseudo-R2	0.205		0.205		0.193		0.159	
AIC/N	1.781		1.783		1.787		1.868	
BIC/N	1.853		1.858		1.826		1.904	
HIC/N	1.808		1.811		1.802		1.882	
#parameters	20		21		11		10	

Note: †Model estimated with uncorrelated parameters.
Standard errors are in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% level, respectively.

Because the two datasets are identical aside from the dependent variable used in estimation (e.g. same 371 respondents but different dependent variable coding, Y vs Yadj, that accounts for or does not account for choice uncertainty, respectively), a pooled variance or paired version of the comparison tests is needed to account for potential differences in WTP estimates. In this case, one can simulate individual-specific WTP values and apply paired comparison tests - such as a Wilcoxon signed-rank test (Hollander and Wolfe, 1999) and the Paired Convolution test (Poe et al., 1997) - which are more appropriate than independent two-sample tests, as they exploit respondent-level correlation structure.

Table H shows the WTP estimates and comparison test results between models with and without choice uncertainty. We estimated WTP in two ways: 1) for the models estimated in WTP-space we follow the same approach as used to compare recreational groups, and 2) for the models estimated in the utility space . where the individual WTP estimates explode to large values, we perform a non-parametric paired bootstrap, taking 10,000 draws, and calculating the mean of the respondents' WTP values by sampling with resample. We estimate the standard error of each mean WTP by drawing 10,000 bootstrap samples and taking the standard deviation of those 10,000 bootstrap-means. Therefore, the mean WTPs estimated by bootstrap for comparison in Table H are different from the WTP estimated by the Delta Method in Table 4.

The WTP estimates obtained under WTP-space models are more stable across specifications than those estimated in utility space. Specifically, the direction and magnitude of differences in WTP between models with and without choice uncertainty are more systematic under WTP-space models – i.e. SQ, MRS and Carbon attributes consistently yield higher WTP values when choice uncertainty is incorporated. For the attributed number of jobs, this occur solely in the RPL-EC specification. In contrast, results from the utility-space models are more heterogenous: incorporating uncertainty increases WTP for some attributes and specifications, for others, it decreases them.

The Wilcoxon signed rank test statistically rejects the null hypothesis that the true median location shift is not greater than zero for all attribute comparisons (except for MRS in GMXL), indicating that WTP values systematically shift with uncertainty. This finding is reinforced by the Paired Convolution test for blue carbon, indicating a full dominance increase in WTP when uncertainty is considered. These results suggest that, rather than simply lowering values, controlling for choice uncertainty tends to increase WTP estimates, especially for blue carbon. This may reflect respondents' higher valuation of blue carbon when response uncertainty is controlled for. Notably, WTP for SQ becomes less negative when controlling for uncertainty in WTP-Space specifications - indicating reduced aversion to policy change - possibly because these specifications capture our assumption that respondents view the SQ alternative more favorably when uncertainty is controlled for.

Table H. Marginal WTP Estimates for Choice Attributes of Parsimonious Linear Models with and without Choice Uncertainty

	Yadj – With Choice Uncertainty Control				Y – Without Choice Uncertainty Control			
	RPL	RPL-EC	RPL-EC 2†	GMXL†	RPL	RPL-EC	RPL-EC 2†	GMXL†
SQ	-19.429^{b}	-373.396^{***b}	15.819^{b}	-105.915^{***b}	-337.079***	866.051***	-110.556***	-238.463***
	(21.058)	(54.720)	(27.956)	(25.364)	(35.418)	(127.260)	(34.964)	(66.756)
MR	1.021^{***b}	-0.890^{***b}	3.363^{***b}	3.683***	3.149***	1.716***	2.051***	3.568***
	(0.207)	(0.299)	(0.418)	(0.487)	(0.148)	(0.637)	(0.393)	(0.456)
Jobs	0.008^{b}	-2.263^{***b}	0.718^{***b}	0.890^{***b}	0.606***	1.019***	0.671***	0.982***
	(0.052)	(0.224)	(0.098)	(0.103)	(0.032)	(0.227)	(0.075)	(0.116)
Carbon	1.132^{***b}	5.418^{***b}	3.397^{***a,b}	2.527^{***b}	2.704***	-6.896***	2.747***	3.296***
	(0.138)	(0.502)	(0.489)	(0.763)	(0.245)	(1.038)	(0.464)	(0.548)

Note: †Model estimated with uncorrelated parameters. Standard errors are in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% level, respectively.
"a" denotes significance of the differences between WTP estimates that incorporate or not choice uncertainty across respective correlated and uncorrelated specifications at the 5% level for the paired convolution test.
"b" denotes significance of the differences between WTP estimates from the two datasets across respective correlated and uncorrelated specifications at the 1% level for the Wilcoxon signed-rank paired test.

These results have important implications for policy design and welfare analysis, as failure to account for preference uncertainty may bias WTP estimates downward, and obscure meaningful heterogeneity in respondent behavior. From a modeling perspective, the results further reinforce the advantages of WTP-space specifications, which yield more stable and interpretable welfare estimates than those obtained from utility-space models.

Appendix D -Posterior cumulative WTP distributions and attribute intensity scenarios

We selected the RPL–EC model in WTP-space (lowest BIC statistic) to collect the 371 posterior-individual attribute coefficients and plot the WTP distributions for each scenario as cumulative (inverse) demand curves. The x-axis shows person-specific draws of WTP for the bundle of attributes and quantities that define each scenario. The y-axis represents the cumulative curves, delineating the proportion of respondents whose willingness-to-pay is at least each corresponding dollar value, i.e. D(p) $= \Pr(\text{WTP} \geq p) = 1 - F_{\text{WTP}}(p)$,[32].

At any fee level p, the height of the $D(p)$ curve yields the predicted adoption/participation rate (analogous to market share). A curve that lies farther to the right therefore represents a stochastically larger WTP distribution—more of the mass of the distribution is concentrated at higher WTP values.

The optimistic scenario (Figure 3) has a wide range of WTP values, showing an inelastic response (small price-sensitivity) until \$600. However, the absolute spread of respondents' WTP estimates for this scenario is highest with a distribution characterized as positively skewed and with a pronounced right tail - indicating that a larger share of respondents expresses relatively high willingness to pay. Roughly 82% of respondents are willing to pay more than \$200 for this scenario, 35% are willing to pay more than \$700, and 10% are willing to pay more than \$800 (Mean=\$518; Median=\$611; SD=275.05; |CV|=0.5314[33]) . Roughly speaking, the optimistic scenario can be priced as high as \$700 per year and still reach a participation rate of more than 35%. For the SQ scenario, the distribution corresponds to the highest absolute coefficient of variation, with higher heterogeneity of preferences (Mean=\$-115, SD=150,|CV|=1.29). The SQ scenario curve also indicates a higher demand elasticity compared with the other two scenarios. According to the SQ graph, approximately 80% of respondents would willingly pay a positive amount to prevent the SQ scenario. The Pessimistic scenario is depicted in the extreme left of the figure, indicating that the probability of a positive WTP is virtually zero. Respondents are price-sensitive in this scenario (Mean=\$-240,SD=\$131,|CV|=0.54).

It is important to point out that the distribution's non-negligible tail suggests that policy design should explore segmentation or tired pricing. Also, the inverse CDFs indicate that the optimistic scenario curve lies consistently to the right of the other curves. Stochastic dominance in this way indicates higher overall consumer surplus[34]. Because most respondents remain supportive of policy changes, even at higher payment levels, policymakers have scope to impose cost-recovery fees or conservation-extra charges with little loss in the participation rate. Moderate, uniform fees could cover basic program costs, and the long right-tail of high-WTP respondents indicates that tiered or distance-based pricing could raise

[32] $F_{\text{WTP}}(p)$ is the empirical cumulative distribution function (CDF) of the posterior individual WTPs. In the CDF, the curve rises from 0 to 1 as one moves rightward along the price (individual WTP) axis; conversely, $D(p)$ falls from 1 to 0 going from the far-left low price to the far-right high price.

[33] This mean of marginal WTP is different than that estimated previously by 2000 draws using the Delta Method. Here, the mean value is calculated as the average of posterior-individual WTPs estimated directly from the RPL-EC model in WTP space. See Greene (2016).

[34] Without an analytical demand function, we compute consumer surplus directly from the empirical WTP distribution. Conceptually, consumer surplus is the integral of the demand curve from the market price (here effectively \$0) up to each individual's WTP. Equivalently, it is the area under the WTP survival function (the proportion of the sample with WTP $\geq$ x) from x = 0 up to the maximum WTP. In practice, since we have each respondent's maximum WTP, the total consumer surplus for scenario j can be obtained simply by summing all individual WTP values for that scenario (assuming the policy is provided free of charge). The average consumer surplus per person is then the mean WTP for that scenario. This approach treats each individual's WTP as the benefit they would receive if the scenario were implemented without requiring payment (Kriström, 1990; Loomis et al, 1998; Boman et al., 1999). This non-parametric approach leverages the empirical distribution without requiring a functional demand form.

additional revenue without markedly reducing overall support. In contrast, maintaining the Status-Quo without compensatory measures will likely face strong political resistance, revealing a latent demand - that is, an unexpressed but sizable public willingness to pay - for upgrades. The pessimistic scenario is even less feasible, imposing even higher opportunity costs - namely, the loss of valued conservation benefits and ecosystem services. Any shift toward the pessimistic scenario would require large subsidies and still attract little public support. Therefore, the public strongly favors expansionary, carbon-efficient marine-reserve designs. The Status Quo carries a substantial opportunity cost in terms of lost social welfare, and any regression toward the Pessimistic profile would require untenable subsidies. Given the stark opposition to the Pessimistic scenario, and the latent willingness to pay for improvement, policymakers should prioritize Optimistic features (larger reserve, higher blue carbon levels, and net job gains) while using moderate scenarios prioritizing one attribute over another (depending upon the trade-offs and opportunity costs) as a fallback if budget constraints bind.

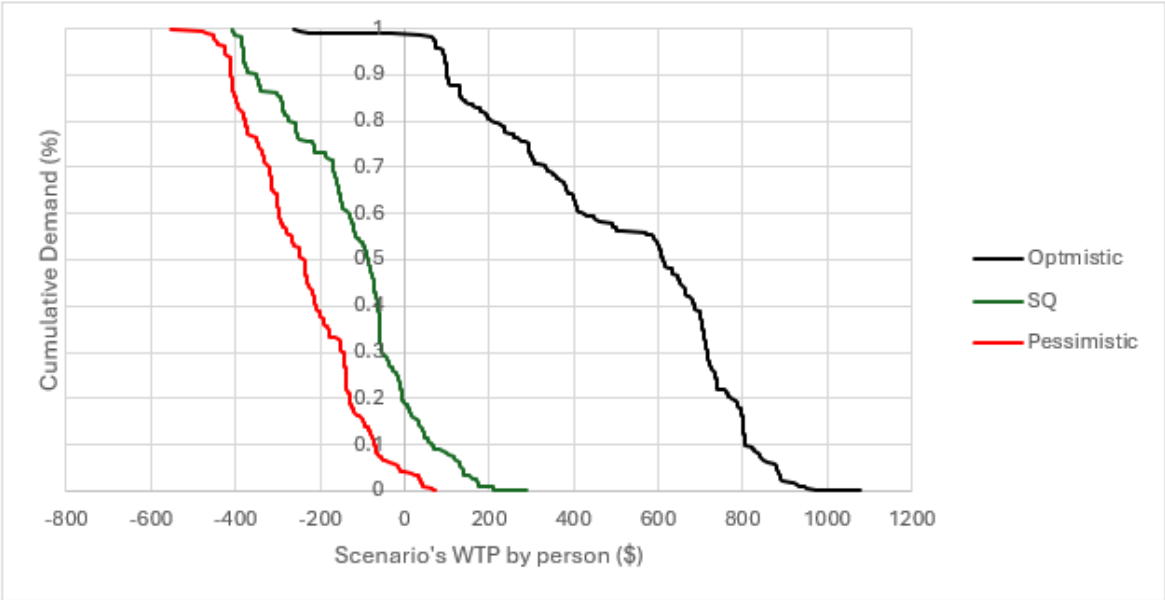


Figure 3: Distribution of WTP for Optimistic, Status Quo and Pessimistic Scenarios

In Figure 4, we build scenarios considering the Optimistic scenario as a baseline. From the baseline, one or two attribute levels are reduced by half, generating scenarios where attributes are prioritized at the expense of others. Thus, we can analyze which attributes should be prioritized when developing policy scenarios in the face of budget constraints, i.e. we can identify which attributes or combinations of attributes generate greater welfare gains and should therefore be maintained when policy makers face budgetary constraints. Comparing the resulting welfare changes pinpoints which attributes - or attribute bundles - yield the greatest marginal welfare gains and thus merit priority when resource constraints are binding.

We define six additional scenarios with various attribute levels compared to the Optimistic baseline: 1) MRCarb-intensive: 50% less net jobs; 2) MRJob-intensive: 50% less blue carbon; 3) CarbJob-intensive: 50% less MR system size; 4) Carb-intensive: 50% less net jobs and MR system size; 5) Job-intensive:50% less blue carbon and MR system size; and 6) Job-intensive: 50% less blue carbon and MR size.

Figure 4 demonstrates that reducing blue carbon by 50% while maintaining high levels of nearshore spatial management size and net job gains results in the largest loss in welfare on average (Mean=$379.59). For each price level more than approximately $140 for this scenario, the percentage of average WTP is lower than any other scenario analyzed. Maintaining high blue carbon along with high levels of jobs

(Mean=$465) or MRS size (Mean=$450) results in similar WTP distributions and means. Until a price of $500, the welfare gains associated with these scenarios are similar to the Optimistic scenario, indicating that, relative to the Optimistic scenario, the opportunity costs up to this price level are insignificant. Beyond this price level, both curves become more elastic, with fewer people willing to pay high taxes for these two scenarios relative to the Optimistic scenario. CarbJob-intensive provides a slightly higher welfare scenario, suggesting that if one attribute must be reduced, it is best to prioritize a scenario including more blue carbon, and at the same time generating a high number of jobs.

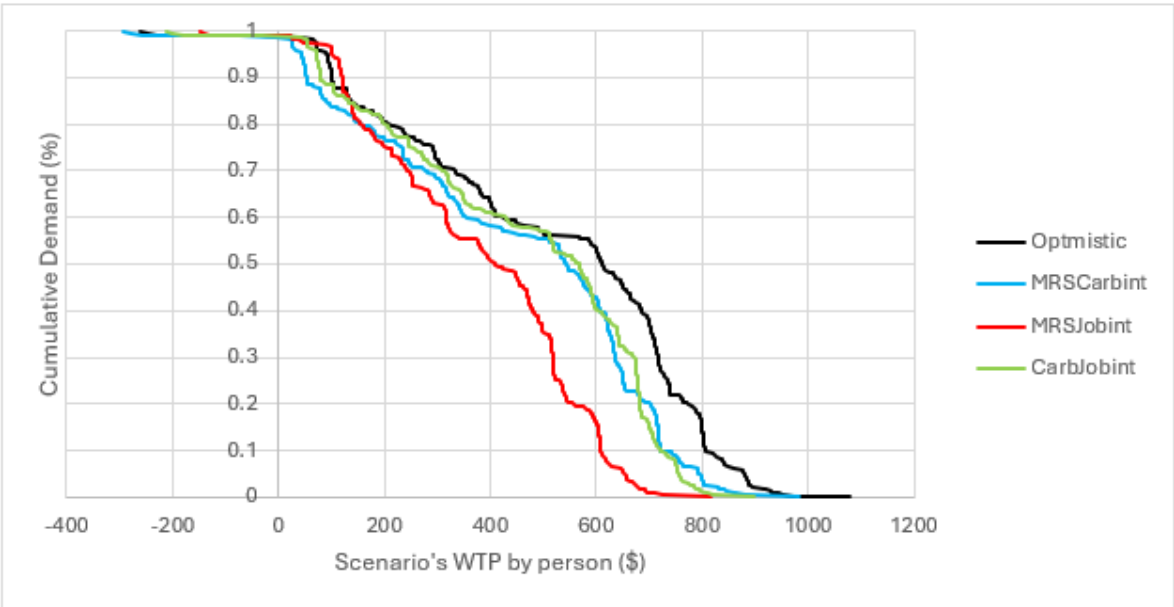


Figure 4: Distribution of WTP for Attribute-Intensive Scenarios (two-attributes intensity)

Figure 5 corroborates the importance of prioritizing blue carbon in a given policy scenario . High levels of blue carbon in the carbon-intensive scenario (Mean=$397) even with 50% lower job levels and MRS size, provides almost the same 60% participation rate as the Optimistic scenario. In this scenario, trading off 50% less net jobs and MR system size while maintaining a higher blue carbon level barely reduces welfare. The mean WTP for blue carbon intensive scenario is higher than in the MRjob-intensive scenario, suggesting that people are willing to trade-off high levels of blue carbon for lower levels of jobs and MRS size. Therefore, it is preferable to prioritize high blue carbon rather than high MR system size or high job increases when the budget is constricted. Furthermore, prioritizing blue carbon provides greater scope for administering fees by tiered or distance-based pricing as a solution to budget constraints. On the other hand, the job-intensive ($326) and the MR-intensive ($311) scenarios result in similar inverse CDFs and welfare gains, yet it is preferable to prioritize net job gains rather than increases in MRS size when the budget is restricted.

Inverse demand curves are a transparent way to communicate adoption forecasts: by simply overlaying the expected program cost and reading off each curve, we obtain the projected uptake rate for each scenario as well as the scenario's cost-sensitivity. The curves can be used to calculate optimal taxes/prices that maximize average net social benefits given average consumer surplus and assumed costs (enforcement, monitoring and administration), where average net social benefit is simply the number of individuals with WTP greater than the tax charged. We can compare distinct average net social benefits across scenarios: for each scenario and the assumed budget estimated per individual, we can construct a vertical line crossing the respective scenario's inverse-demand curve and calculate the are

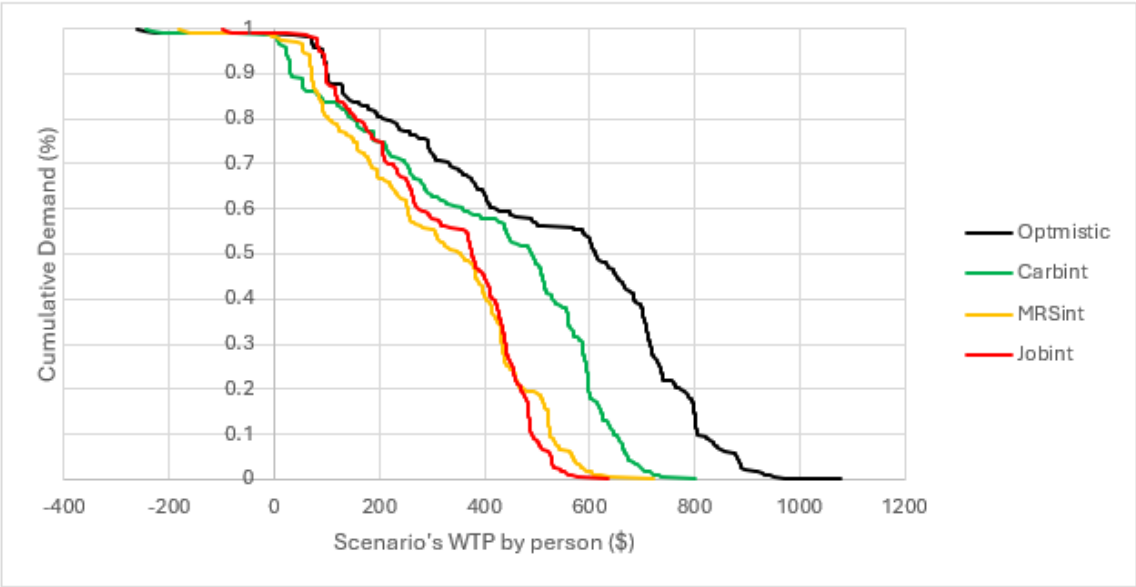


Figure 5: Distribution of WTP for Attribute-Intensive Scenarios (one-attribute intensity)

under the curve from the right of the estimated average assumed costs. Assuming that the tax will be charged uniformly across individuals, the most preferred scenario has the highest difference between mean WTP and average cost[35].

Table I. WTP Estimates for Attribute-Intensive Scenarios

RPL-EC -2 (WTP-Space)†			
CarbJob-int	460.230***	Carb-int	393.096***
	(59.826)		(55.627)
MRCarb-int	444.381***	MR-int	307.043***
	(62.593)		(41.275)
MRJob-int	374.178***	Job-int	307.043***
	(46.075)		(41.275)

Note: WTP's scenarios with estimated standard errors using the Delta Method.
†Model estimated with uncorrelated parameters.
Standard errors are in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% level, respectively.

[35]In this case, when we talk about net social benefits (NSB) we are including as benefits solely the estimated consumer surplus derived from the scenario's attribute levels. A more complete NSB also includes producer surplus estimates derived from harvest profits and other opportunity costs related to fisheries.

Table I shows mean WTP values for the distinct attribute-intensive scenarios, using the coefficients from the RPL-EC model estimated in WTP space (which were used to derive the posterior individual WTP values comprising the inverse CDFs of Figures 3 to 5). These values corroborate the conclusions drawn from the figures. We again see clearly how blue carbon is the attribute which should be prioritized in our policy scenarios. All scenarios with high levels of blue carbon possess higher mean WTPs compared to the other scenarios that have high levels of MRS size and/or net jobs.

Appendix E

Table J. Full attribute attendance (FAA) results — coefficient estimates.

Attribute	**Coefficient Estimate**
Latent Class 1 (23.83%)	
SQ	-1.60^{***} (0.27)
MR	0.02^{**} (0.005)
Carbon	-0.02^{*} (0.006)
Jobs	0.004^{**} (0.004)
Cost	-0.002^{*} (0.001)
Latent Class 2 (53.19%)	
SQ	-1.37^{**} (0.20)
MR	0.005^{**} (0.001)
Carbon	0.01^{**} (0.001)
Jobs	0.02^{**} (0.001)
Cost	-0.001^{**} (0.0004)
Latent Class 3 (22.99%)	
SQ	-1.17^{***} (0.35)
MR	0.04^{**} (0.01)
Carbon	0.03^{**} (0.008)
Jobs	0.005^{**} (0.002)
Cost	-0.02^{**} (0.004)
log-likelihood	-1,305.05
χ^2	664.12^{***}
pseudo-R^2	0.20
AIC	1.782
BIC	1.842
HIC	1.804

Notes: Standard errors are in parentheses. *** significant at 1%, ** at 5%, * at 10%.

Table K. Attribute non-attendance (ANA) results — coefficient estimates.

Attribute	**Coefficient Estimate**
Latent Class 1 (16.41%)	
SQ	-2.28^{***} (0.33)
MR	0.01^{***} (0.001)
Carbon	0.01^{***} (0.001)
Jobs	0.002^{**} (0.0002)
Cost	-0.006^{**} (0.0006)
Latent Class 2 (50.83%)	
SQ	0.0 (—)
MR	0.01^{***} (0.001)
Carbon	0.01^{***} (0.001)
Jobs	0.002^{**} (0.0002)
Cost	-0.006^{**} (0.0006)
Latent Class 3 (32.77%)	
SQ	-2.28^{**} (0.35)
MR	0.01^{***} (0.001)
Carbon	0.01^{***} (0.001)
Jobs	0.002^{**} (0.0002)
Cost	0.0 (—)
log-likelihood	-1,401.63
χ^2	470.96^{***}
pseudo-R^2	0.14
AIC	1.898
BIC	1.923
HIC	1.908

Notes: Standard errors are in parentheses. *** significant at 1%, ** at 5%, * at 10%.

Appendix F

Table L. RPL Linear Model Results with Fisherman's Interactions by Subsamples Related to General Recreation in the Oregon Coast

	General Recreation (n=230)		**No-General Recreation (n=118)**	
	RPL 1†		RPL 2†	
Attribute	Mean	SD	Mean	SD
SQ	0.571	2.104***	1.006**	2.733***
	(0.354)	(0.267)	(0.449)	(0.503)
MR	0.018***	0.012***	0.014***	0.016***
	(0.004)	(0.004)	(0.004)	(0.005)
Jobs	0.004***	0.003***	0.005***	0.002
	(0.0008)	(0.0009)	(0.001)	(0.002)
Carbon	0.022***	0.014***	0.016***	0.022***
	(0.004)	(0.004)	(0.005)	(0.007)
Cost	-0.005***	0.005***	-0.006***	0.008***
	(0.0008)	(0.001)	(0.002)	(0.002)
SQ*Fish	-0.126		0.255	
	(0.418)		(0.742)	
MR*Fish	-0.003		-0.011	
	(0.004)		(0.007)	
Job*Fish	0.00007		-0.003**	
	(0.0009)		(0.0014)	
Carb*Fish	-0.015***		-0.002	
	(0.005)		(0.007)	
Log-likelihood	-819.331		-406.222	
Pseudo-R2	0.190		0.231	
AIC/N	1.812		1.781	
BIC/N	1.885		1.904	
HIC/N	1.840		1.829	
#parameters	14		14	

Note: †Model estimated with uncorrelated parameters.
Standard errors are in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% level, respectively.

An interesting way to look the effect of being a fisherman on preferences is by analyzing fisher's interaction with each attribute (Table L). Results show that among people who engage in general recreational activities, also being a fisher reduces an individual's utility level associated with blue carbon, suggesting that fishers care less about climate change and may perceive carbon-related restrictions as conflicting with their recreational fishing practices At the same time, individuals in the "No-General Recreation" subgroup who engage only in fishing-related recreational activities tend to value ocean job-related increases less than those who do not engage in any form of recreation. This possibly indicates greater skepticism or a lower perceived benefit from ocean-related employment opportunities among those who engage solely in fishing-based recreation, compared to individuals who do not participate in

any ocean-related recreational activity. In essence, the results indicate that recreational fishermen are not a monolithic group. Their valuation of nearshore spatial management policies depends on whether they also engage in broader ocean recreation or solely fish - which reveals layered, context-specific trade-offs that simple subgroup labels potentially obscure. Appendix B shows MNL results for subsamples of people who do not participate in fishing recreational activities plus engage or not engage in general ocean recreational activities.